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Thailand's Car Tax Rebate Scheme and Consumption Responses: the Role of Durable Goods with Adjustment Costs

by

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Abstract

In 2011, Thailand faced the largest flood in seventy years. In response to the unexpected crisis the Thai government rolled out Thailand's car tax rebate scheme in an attempt to prevent the economy from slipping into a deep recession. This study investigates consumption responses to changes in vehicle prices induced by the car tax rebate scheme presented in the framework of a life-cycle model. The model features durable goods with adjustment costs and non-homothetic preference. The key features match the fact that car purchases are lumpy and infrequent and that cars are luxury goods in Thailand. Additionally, liquidity constraints and adjustment costs are also important features for the evaluation of shorter-run consumption responses. Key parameters are estimated to match household-level data. Then partial equilibrium responses, which are key inputs to inform the aggregate outcome of the policy, are simulated given a distribution of the population wealth, income, and age in the economy. Findings show that Thai households have large elasticity of intertemporal substitution (EIS), hence large responses to the fiscal stimulus. Furthermore, non-homotheticity in the preference generates heterogeneous policy responses varied by household income and wealth. The model predicts that the temporary price shock will lead to a large cutback in future consumption and saving, consistent with the evidence shown by aggregate data. A number of alternative policy experiments are also conducted.

1 Introduction

Many experts criticized the car tax rebate program for its unintended effect of dampening the economy by putting households in debt hence lowering household consumption after the scheme was over. The private consumption expenditure (PCE) were boosted tremendously during the policy and slowed for many consecutive quarters afterwards. Furthermore, The car sales data suggests that the future demand was pulled forward during the program, and cannibalized new car sales in the following years. The excise taxes of as many as 1.1 million cars and pick-up trucks were rebated in the scheme. A naive linear projection in car sales suggests a conservative approximation of additional 600 thousand out of 1.1 million vehicles were being purchased during the policy (Figure 3).

While evidence from aggregate data is pointing towards the dampening effect of the policy on future household consumption, to the author's knowledge no formal study or evaluation of the impact of the policy on household consumption responses has been conducted, or key parameters that are crucial to the size of household responses such as elasticity of intertemporal substitution (EIS) have been measured. Primary evidences from aggregate data suggest large EIS among Thai households. That is households were willing to shift future consumption forward to purchase vehicles during the policy in responses to the unanticipated price shock. Nonetheless, the conclusions that can be drawn from the aggregate evidence is limited due to many confounding factors that may contribute to the slowdown, including lower income due to the recession and flood, and tepid global economy. This study fills that gap by proposing a model that can explain household car-purchasing decisions, estimating model parameters to match Thai household preferences and Thai economy, providing a counter-factual to the policy, and conducting alternative policy experiments.

The car tax rebate affected both household current and future consumption by operating through a number of channels. Automobiles were unexpectedly and temporarily cheaper relative to non-durable goods and asset returns, and hence became more attractive as both assets and consumable goods. The tax break influences household consumption decision via 1) substitution effects due to the changes in relative prices

that lead to both intratemporal and intertemporal substitution in a dynamic setting; 2) income effects as cars become less expensive and demand for cars and other goods weakly increases; and 3) wealth effects since the tax break may influence car resale prices and cars also function as illiquid assets or buffer stocks for households. In the presence of a price shock, households may intratemporally shift their consumption away from other types of consumption and intertemporally shift their future consumption towards car demands. This study will propose a life-cycle model that captures all of the abovementioned channels that the car tax rebate scheme operated through.

To evaluate the policy impact and provide a framework for future policy evaluation, this study proposes a heterogeneous agent model with two goods, non-durable and durable, with an emphasis on the dual roles of durable goods as 1) long-lived consumable goods and 2) assets that store wealth and depreciate over time. Furthermore, non-durable goods are a necessity while durable goods are a luxury, matching the context of the Thai economy. The framework of heterogeneous agents in the life-cycle model also allows for the evaluation of the policy on households at various socioeconomic status, including age, income and wealth. In the model, households will face uninsurable income risks, liquidity constraints, and adjustment costs of durable goods. The presence of adjustment costs and liquidity constraints is essential for the evaluation of the shorter run consumption responses.

The extent to which households will respond to a tax rate change depends on two key parameters: elasticity of intratemporal substitution, and elasticity of intertemporal substitution. These key parameters are not only important for the evaluation of the car tax rebate program but also quantitatively informative for future economic policy decisions. In this study, the model parameters are estimated to match a rich panel of micro-level data, the Townsend Thai data, and EIS for Thai households are calculated. Model parameters are estimated using Generalized Method of Moments (GMM) to match moments of assets, consumption, rates of car ownership, and the probability of durable adjustment from household-level data.

The estimation places an emphasis on relevant parameters to quantify EIS of Thai households. Additionally, the model generates heterogeneous EIS and consumption responses to the car tax rebate among households. The varying EIS among households

with different levels of wealth and income are generated from the income effect given the non-homothetic preference. Low income (wealth) households do not own a car, and only do so when they reach sufficiently high level of income (wealth). Relevant parameters in the structural model are estimated and heterogeneous responses at various income, wealth, and age levels are numerically calculated from simulations. The results are consistent with recent works that use GMM approach that found EIS to be well above 1 [Hansen et al., 2007, Barro, 2009, Bansal et al., 2012, Gruber, 2013].

This study proceeds as follows: Section 2 presents related literature, including studies on similar tax rebate policy and consumption of durable goods, and literature on EIS estimations. Section 3 reviews Thailand's car tax rebate policy and aggregate data that motivates the research question and modeling choice. Section 4 explains the model, and discusses the model solution, computation and interpretations of model parameters. Section 5 discusses estimation strategy by first explaining the Thai household data, parameter choices, estimation method, and identification. Then Section 6 presents estimation results, model fit, and EIS estimates. The counterfactual and policy experiments are presented in Section 7. Alternative policy experiments, including a consumption tax break and income tax break, are conducted. Lastly, Section 8 concludes the study.

2 Literature Review

2.1 Tax Rebate Studies

Durable spending is one of the first GDE components to drop during a downturn, making durable stimulus packages a popular fiscal tool during recessions. A car tax break pulls future demand forward to when the market is supposedly operating inefficiently at below the optimal level. Temporary tax cuts or subsidies on personal vehicles potentially lead to a surge of durable spending during the program at the cost of cannibalizing future purchases [Mian and Sufi, 2012]. However, the sizes of the reversal effects of the policy on future demand have been mixed. Key parameters influencing the size of the effects are the elasticity of intratemporal substitution and the elasticity of intertemporal

substitution.

Thailand's first car tax rebate program is similar to other subsidies programs previously implemented in other countries. For instance, the US's Cars Allowance Rebate System (CARS), colloquially known as Cash for Clunkers, and the France's Balladurette and Juppette's program which subsidized a replacement of old cars with new ones. However, Thailand's car tax rebate program is different from those previous subsidies in that to receive the rebates, buyers do not need to trade in an old car. This is because the main purpose of Thailand's car tax rebate scheme was not an environmental one. Its primary goal was to boost domestic production and consumption via tax incentives. Since the stock of old cars were never destroyed as required by CARS or Balladurette and Juppette, Thailand's car tax rebate program has large general equilibrium effect on car prices. Furthermore, the size of the program was significantly larger than other previous programs due to the high excise tax rate on cars in Thailand. Previous comparable programs were shown to have different effects. While Balladurette and Juppette subsidies increased government revenues in the short run but cannibalized the long-run revenues to lower than the baseline level without the intervention [Adda and Cooper, 2000], the Cash for Clunkers program appeared to boost short-run aggregate demand with no or inconsequential effects in the long run [Mian and Sufi, 2012]. Mian and Sufi [2012] estimated that Cash for Clunkers program pulled future demand from a very near future of only about 7 months, and found no evidence of the policy impacts on employment, house prices, and household default rates.

Aside from being related to literature on car subsidies, this study is also relevant to literature on tax rebate and household consumption responses. There are many recent empirical works on income tax rebate and consumption responses, Shapiro and Slemrod [2009], Parker et al. [2013] to name two. However, the car tax rebate operates through different channels than income tax rebate. This is due to the dual roles of durables as consumable goods and illiquid assets. Recent studies that focus on household consumption responses with the presence of durable or illiquid assets include Kaplan and Violante [2014], Berger et al. [2018], Berger and Vavra [2015].

A permanent income hypothesis (PIH) and rational expectation would predict households to respond to the price shock by adjusting their durable stock and smooth

their consumption at the arrival of the information about the scheme. However, due to liquidity constraints and transaction costs, household consumption responses could be different from the theory prediction. Among the first studies to examine PIH for durable goods, Bernanke [1984] used panel data and incorporates durable consumption to test the joint hypothesis. The result does not reject PIH. It concluded that there is no evidence that liquidity constraints or capital market imperfections are important in the timing of family car purchases as households can plan durable purchases long in advance. Yet the result does not refute the fact that Thailand's car tax rebate scheme could substantially change the timing of household durable purchases. The policy introduced an unexpected and temporary, yet very sizable, price shock to the car market. Liquidity constraints could become an important factor since the change was unanticipated. Later works emphasize the importance of adjustment costs in household durable adjustment due to a few observable facts about durable adjustment. First, households only occasionally adjust their stock of durable goods [Lam, 1991]. Moreover, consumers face lumpy transaction costs, and either fully adjust by replacing their old durable goods or do not adjust at all [Bar-Ilan and Blinder, 1992]. Such behavior is consistent with a threshold rule or more generally the (S,s) rule [Caballero and Engel, 1999], in which consumers adjust their durable stocks only when their actual stock deviates from the optimal stock by a critical amount.

Many recent works also confirm the importance of liquidity constraints on the short and medium term consumption responses. Assuming nonseparability in preferences between durable and nondurable goods, intratemporal substitution between durables and nondurables will also lead to substantial intertemporal substitution. Previous studies have shown that due to the intratemporal and intertemporal substitution between durables and nondurables, consumption pattern and investment in assets in the short term and medium term predictably will change especially for those more credit constrained [Cerletti and Pijoan-Mas, 2012]. The transmission of income shocks could be even larger among the constrained households.

Unlike an income tax rebate, only households that adjust their car stocks can directly benefit from the program. Those who benefited from the first car stimulus package must be able to afford a car and have access to capital at the time. Liquidity

constraints become an important determinant of the access to the scheme participation. Furthermore, due to the fact that personal vehicles are luxury goods in Thailand, the presence of income and wealth effect will also be a critical feature to the model

2.2 EIS Literature

As mentioned before, EIS is a key parameter that would quantify the impact of the car tax rebate on future consumption. EIS generally measures the willingness of consumers to substitute between periods, given a change in prices of consumption or asset returns. It is a crucial input to determine household responses to various fiscal stimuli. This study fills a gap in the literature by focusing on estimating EIS in a structural model with adjustment costs, and non-homothetic preference in durables.

There has been a wide range of EIS used and estimated in the past. Estimates of EIS vary across countries, wealth levels and depend upon an inclusion of durable goods in data. In the macroeconomic literature empirical studies using aggregate consumption data typically find EIS to be close to zero. On the other hand, models that are designed to match growth and fluctuation generally would need EIS to be close to one [Guvenen, 2006].

The wide EIS range among countries can be reconciled with two main types of heterogeneity: stock market participation and wealth. Blundell et al. [1994] and Attanasio and Weber [1995] suggest that rich households tend to show a larger EIS. Furthermore, households with high wealth– generally also stockholders– are found to have high EIS while households with low wealth are found to have lower EIS [Guvenen, 2006, Havranek et al., 2015]. Havranek et al. [2015] found that countries with low participation in stock markets display a smaller value of elasticity. The mean estimates reported in empirical studies is 0.5 [Havranek et al., 2015].

The inclusion of durable goods also significantly affects EIS estimates. Early and influential work by Hall [1988] found EIS to be close to zero indicating that consumption changes very little in response to a change in real interest rates. Mankiw et al. [1985] enriched the model by introducing the service flow from consumer durables with a linear Engel curve in the model and found EIS for durable goods close to one. Ogaki

and Reinhart [1998] used the innovative cointegration method, which is robust to the adjustment costs, and assumed non-separability between non-durables and durables. They estimated EIS to be around 0.32-0.45 with CES utility function specification. Recent work by Havránek [2015] extends the work of Ogaki and Reinhart by incorporating non-homotheticity for durable goods in order to account for the potential income effect. He found that EIS of total consumption to be small (weighted average of the consumption basket with negligible EIS for nondurables and significant EIS for durables).

A long line of literature has acknowledged the key role that adjustment costs play in consumption of durable goods [Bernanke, 1984, Eberly, 1994, Grossman and Laroque, 1987, Lam, 1991]. Adjustment costs and liquidity constraints will play a key role for work interested in shorter run consumption responses. Recent and prominent works that model durable consumption or illiquid assets with transaction costs [Berger et al., 2015, Kaplan and Violante, 2014] have refrained from estimating the EIS parameters and assume a common value of parameters in the literature. This study contribute to the EIS literature by incorporating adjustment costs in the structural model and focus on estimating the EIS parameters.

The key parameters that influence EIS in the context of the model in this study are the curvature of the utility function and non-homotheticity. These two parameters together with the durable adjustment costs in the model generate heterogeneity in EIS and non-linear household consumption responses. The model has features that match the empirical evidences that car purchases are infrequent and lumpy [Lam, 1991, Eberly, 1994], and that the Engel curve of durable consumption is nonlinear [Havránek, 2015].

Due to the rich features of the model there is no closed-form solution for EIS. For a model with stochastic income, the literature has mostly used the Euler's equation method to estimate EIS rather than estimating relevant parameters in a model directly [Attanasio and Weber, 2010]. The Euler's equation is robust to the presence of adjustment costs, albeit only the long run. However, the shorter run responses are of great interest to policy makers and are affected by the adjustment cost. Instead of the Euler's equation method, this study uses a direct parameter estimation from a structural model.

More recent works with GMM approach to estimate EIS instead of a regression

approach found the elasticity to be above 1 [Hansen et al., 2007]. Another strand of literature that estimates EIS is the asset pricing literature. Two recent approaches, the long-run risk hypothesis [Bansal and Yaron, 2004] and the rare disasters model [Barro, 2009] showed that EIS needs to be above 1 to replicate consumption volatility and asset prices in data. Bansal et al. [2012] discusses that the low estimates of EIS from the regression method could be severely downward biased due to measurement error in the real rate, or endogeneity bias from omitted variables that correlate with consumption volatility or the real interest rates. Gruber [2013] address endogenity bias by exploiting exogenous differences in capital income tax rates and found estimates to be around 2.

3 Policy Review

3.1 Overview

The flood in 2011 took Thailand by a surprise. Not only residential areas were damaged by the flood, but also were the agricultural and industrial areas. Thailand has the largest automobile manufacturing base in the Southeast Asia, hence the Thai government stepped in to brace automobile manufacturers as well as to stimulate domestic spending by offering tax benefits to new car buyers.

The car tax rebate scheme offered new car buyers the vehicle excise tax rebates after one year of ownership of qualifying vehicles. The rebate amount is equal to the vehicle excise taxes, but not exceeding 100,000 Baht (approximately 2,900 USD at 1USD to 35 Baht exchange rate). The vehicle excise tax rates varied depending on the types and models of vehicles, ranging from 3 to 25 percent. Specifically, the excise tax rates were 3 percent for two-door pick-up truck, 12 percent for four-door pick-up trucks, 17 percent for eco-friendly passenger cars and 25 percent for small size passenger cars. Qualifying vehicles must be manufactured in Thailand, cost less than 1 million Baht, and the engine size not exceeding 1500 cc for passenger cars. The program opened for interested participants to enroll between October 2011 and 31 December 2012. However, due to high excess demand, the actual vehicle disbursement and tax rebate reimbursement lasted into 2014.

The size of the program was enormous, far exceeding the government's initial expectation. Over 1.25 million Thais signed up between September 2011 and December 2012, and in the end as many as 1.1 millions personal vehicles were purchased and excise taxes reimbursed from the scheme. Auto sales hit the record high in 2012 and 2013 during the scheme. The Thai economy was seen at a growth of 7.3 percent in 2012, climbed up from a bottom of 0.8 percent growth in 2011.

From a fiscal standpoint, the program adversely affected the government's budget balance. According to a study by Thai Parliamentary Budget Office, the government may lose approximately 20-30 billion baht from this scheme (TDRI, 2014). The drop in car sales following the scheme led to a loss in tax revenues while ongoing tax returns imposed extra budget burden.

3.2 Car Prices and Resale Market

The policy also put a downward pressure on used car prices (Chaithongsri, 2013). Lower new car prices and the rise in used car supply depressed the resale values. According to the Bank of Thailand, some model-year car prices dropped down below the debt value that the cars was mortgaged for. Consequently, many debtors defaulted and let their cars go to purchase a new one at lower prices. The Bank of Thailand reported that the level of non-performing loans (NPL) rose from 1.3 percent in 2012, at the beginning of the scheme, to 2.5 percent in 2014 (BOT, 2016).

To monitor the second-hand markets, the Bank of Thailand also started constructing a used car price index (UCPI) in 2011. Figure 8 shows a summary of car prices from various sources, including Consumer Price Index (CPI) Vehicle, the Bank of Thailand's UCPI (on the left axis) and the trend of household expenditure on car purchases from Townsend Thai Data (right axis). The trend of price indexes and household spending per vehicle illustrate the general equilibrium effect of the policy on new and used car prices. The used car prices started to drop immediately during the policy, hit a low in 2014. It dropped by approximately 25% from its peak in 2012. The CPI, which is comprised of only new car prices, also fell by approximately 20% following the policy.

The average household spending on a newly acquired vehicle in Townsend

Thai Data from 2005 to 2015 is shown in Figure 8. The reported personal vehicle spending consists of spending on both new and used cars. The average spending per vehicle roughly follows the trend of CPI prior to and during the policy. Following the policy, household durable spending per vehicle dropped drastically and largely tracked UCPI. This suggests that most vehicle purchases households made during the policy were on new cars, while most purchases households made after the policy were used cars. Households substituted away from new cars to used cars as the used cars prices plummeted. However, in 2015, the new car prices dropped to compete with used car prices and household spending per vehicle bounced back again due to the drop in new car prices.

3.3 Vehicle Production and Sales

As many as 1.1 million personal vehicles were purchased and reimbursed for the tax rebate. Thailand's total annual car sales in 2012 and 2013 were 1.4 million and 1.3 million in comparison to 800 thousands and 795 thousands in 2010, and 2011 (Figure 1), an addition of approximately 600 thousands and 500 thousands in 2012 and 2013 from 2011. Both pick-up and and passenger car sales surged by about the same amount during the rebate. (Figure 1). Most of the surge in car sales was likely a result of the car tax rebate program rather than other confounding factors given that the macroeconomic conditions were during a downturn.

Car production data shows that the surge in car purchases during the policy cannibalizes future new car sales, and the policy led to bear markets for both eligible and non-eligible vehicles following the policy. Figure 2 shows domestic car production by rebate eligibility. The production of eligible vehicles was significantly higher, confirming that most of the surge in car purchases was the result of the policy. Total non-eligible automobile production steadily increase up to 2012, and dropped in year 2013, and 2014. The number seemed to recovered in 2015 but not to the previous level as before the policy. Meanwhile, the production of eligible vehicles surged during the policy, and dropped dramatically after. There is also a downward trend in both eligible and non-eligible car production following the policy.

3.4 National Accounts Data

Besides the effect on automobile markets, a stimulus of such a large scale will likely have significant impacts on the wider economy. This section focuses on the policy impact on the demand side by examining gross domestic expenditure (GDE) and private final consumption expenditure (PFCE or PCE).

This section investigates the effect of the policy on future aggregate consumption by examining Private Consumption Expenditure (PCE) data. Personal vehicle purchases growth in PCE was negative for as many as eleven quarters following the policy (Figure 4). Assuming no substitutibility and complementarity of car purchases on other spending, PCE would be predicted to be lower due to future demand being pulled forward during the policy (Figure 3.)

To investigate substitutibility (or complementarity) between various types of spending, PCE data aggregated by durability is also examined. The aggregate household consumption expenditure in non-durable, semi-durable, and services surged with car purchases during the scheme. The aggregate trend suggests that durable and non-durable spending may be complementary. However, as private vehicle purchases decreased dramatically after the program, growth of household expenditure in non-durable and semi-durable also declined with the durable spending. This suggests strong elasticity of intertemporal substitution among Thai households that are willing to shift future consumption forward due to the change in car prices.

Aggregate data shows evidence of a large elasticity of intertemporal substitution among Thai households, suggested by the enormous surge in household spending during the policy and a relatively long spell of sluggish consumption, even negative growth in some quarters, for some years following the policy. Growth of private vehicle purchases was negative for as long as eleven consecutive quarters after the policy ended in 2013 (Figure 4). PFCE moved closely with personal vehicle purchases, and had negative growth for 3 quarters after the policy ended (Figure 5). In fact, aggregate domestic consumption remained slow for at least three following years. Figure 5 also shows Domestic PFCE which refers to PFCE before adding expenditure of residents aboard and subtracting expenditure of foreigners in the country. Improved GDE at the

end of 2014 was from the acceleration of Domestic PFCE which included spending of foreigners in the country. However, PFCE, foreigner expenditure excluded, continued to decelerate until the end of 2015. This suggests that the recovery of GDE in 2013 and 2014 was contributed by tourist spending while domestic purchasing power continued to be weak.

Figure 6 shows Domestic PFCE broken down by durability. These are components of Domestic PFCE, therefore expenditures by tourists in the country are included. The share of durable spending is approximately 10% (Figure 6 from 2005 to 2015, and it surged during the policy and quickly subsided afterwards. Yet the level of durable share after the program ended seem to be slightly higher than prior to the program. This could be due to the general equilibrium effect that the policy permanently reduced the user costs of personal vehicles.

During the program, spending on semi-durable was the least affected, whereas shares of non-durable and services were smaller during the policy. After the policy, share of spending on services resumed to the level before the policy, while the share of non-durable spending continued to be smaller. Most of tourist expenditures fall in the service category, i.e. spending on hotels, restaurants, and transportation services. This suggests the that the recovered aggregate spending was mostly boosted by foreign tourist spending.

Figure 7 shows growth rates of domestic PFCE by durability. While durable spending accounts for the smallest share in expenditure, its growth is the most volatile. Growth of semi-durables and non-durables in 2012 and 2013 seems to move in the same direction as growth of vehicle purchases and durables. This could be due to the fact that the income effects from cheaper cars were larger than the within-period substitution effect during the policy. Consumers might be spending more on complementary goods to personal vehicles during the policy due to the perceived rise in real income from the tax break. However, following the policy durable spending declined drastically and spending in other categories also dropped. The exception was spending on services that recovered in the fourth quarter of 2013, which again was likely propelled by the recovery of the tourism sector.

In conclusion, aggregate data suggests that domestic spending was significantly boosted during the policy. Spending in all categories increased likely due to the rise in real income from the tax break. However, domestic demand remained weak for a few years afterwards. The following recovery of private consumption was most likely contributed by external factors rather than domestic.

An overview at the aggregate data confirms that Thailand's car tax rebate had a significant impact on household consumption, asset holdings and saving. However, to disentangle the impact of the policy from other confounding factors, such as the looming recession at the time or low income due to slow global economy, a more in-depth analysis is required. The next section develops a life-cycle consumption and saving model with cars as durable goods in order to formally evaluate the policy impact on household consumption responses.

Given the survey primary data, this section concludes by proposing features that are important for a model to realistically evaluate shorter run consumption responses to the car tax rebate policy. Namely, the model should aim at estimating elasticity of interetmporal substitution and allowing for intratemporal substitution between durables and nondurables. Furthermore, the model should also include stochastic and uninsured income uncertainty, liquidity constraints, adjustment costs in durable goods. Additionally, durable goods should be modeled as a luxury good to fit the context of the Thai economy.

4 Model

Households consume two types of goods: non-durables and durables, accumulate risk-free assets as well as durable goods, and are able to borrow within borrowing constraints. Durable goods serve dual roles: providing a stream of services, and storing wealth. For the purpose of evaluating Thailand's car tax rebate policy, durable goods in this study are only comprise of household personal vehicles, including passenger cars and pick-up trucks (henceforth referred to as cars for brevity).

In other related studies, durable goods generally include housing, personal vehi-

cles, and other board durables. The inclusion of data depends on the research question. This study includes only cars as household durable goods with incurred adjustment costs to evaluate the car tax rebate policy and estimate relevant parameters for Thai household preferences with respect to cars. However, in order to accurately capture the wealth of households, household assets in the model consist of both household liquid and fixed assets. The majority of household fixed assets are in the form of land and housing (Table 2). Relative to cars, wealth stored in land and housing is much more accessible. Land and housing can be liquidated by collateralization relatively easily given the model period of one year. Therefore, the inclusion of all household fixed assets, besides cars, as household wealth that can be accessed without adjustment costs is fairly innocuous.

Liquidity constraints and adjustment costs are crucial in generating shorter-run responses to unexpected price changes. The adjustment costs of personal vehicles in the model encompasses relevant costs in resaling, such as the time cost, registration fees, and asymmetric information costs. Without the adjustment costs, households will adjust their cars every period and all the car owners would have participated in the car tax rebate policy. Liquidity constraints are another important friction for the car adjustment, so that households without sufficient asset or income cannot purchase a car due to limited access to credit.

4.1 Setting

Households live for 60 periods, $j = 1, ..., 60$, representing households age of 26 to 85. This is measured by the age of the head of the household in data. They consume two goods: non-durable goods and a stream of services from durable goods, C_{it}, and D_{it} respectively in each period while receiving exogenous stochastic income of Y_{it}. They also store their wealth in two assets: a risk-free asset with no adjustment costs, A_{it}, and durable goods, D_{it}. A household maximizes the expected utility function

$$E_t[\Sigma_{j=1}^{J}\beta^j U(C_{i,t+j}, D_{i,t+j}) + \beta^{J+1}B(W_{it+J+1})]$$

where $B(.)$ is the value of terminal wealth in period 60.

4.1.1 Preferences

In each period, a household consumes two types of goods, non-durable goods and service flow from durable goods. The stream of service flow is proportional to the size of durable, consistent with the standard assumption in the literature. Cars are luxury goods in Thailand, so the preference is assumed to be non-homothetic in car consumption to capture the non-linear Engle curve in durable spending.

When EIS is the main research interest, non-separability in durable goods and non-durable goods could be a concern. If the intratemporal substitution effect between durable and non-durable is strong, assuming additive separability between them will lead to a bias in estimation of EIS. The standard practice, pioneered by Ogaki and Reinhart [1998], is to assume the standard CES functional form and estimate the intratemporal substitution parameters. This study chooses the Cobb-Douglas utility function, which is equivalent to the CES utility function with the intratemporal substitution of 1, as a benchmark for two reasons. First, the estimates of elasticity between non-durables and durables in the literature are not statistically different from 1 [Berger and Vavra, 2015]. Second, the Cobb-Douglas utility function is chosen as a benchmark to be consistent with the most recent works in the literature on household durable goods with adjustment costs [Berger and Vavra, 2015, Berger et al., 2015, Fernandez-Villaverde and Krueger, 2011].

Households are assumed to have a per-period Cobb-Douglas utility function as follows:

$$U(C_{it}, D_{it}) = \frac{1}{1-\sigma}(C_{it}^{\alpha}(D_{it}+\tau)^{1-\alpha})^{1-\sigma}$$

Non-homotheticity in the preference distinguishes this study from closely related studies. In a static setting with no asset accumulation and deterministic income, the share of durable and nondurable goods demand is constant for any given level of income. In other words, there is no income effect. Adding the τ parameter in the utility function allows for the income effect in durable goods: allowing the income expansion path to be non-linear in cars. Households with sufficiently low income (or wealth) will

have the share of nondurable to durable goods at 1:0. When household asset holding or income is sufficiently high, the ratio of $C : D + \tau$ is equal to $\alpha : 1 - \alpha$. As income rises, the ratio of $C : D$ will converge to $\alpha : 1 - \alpha$. In a dynamic setting, the ratio of $C : D$ could be different as governed by the intertemporal and intratemporal marginal utility conditions.

With the Cobb-Douglas utility function specification, the intratemporal substitution parameter is assumed to be equal to 1. However, with non-homotheticity in the preference, the intratemporal substitution varies among households according their wealth and income level relative to the non-homotheticity parameter. Low-income (wealth) households have relatively low intratemporal substitution, holding real income constant they may not change their durable consumption as relative prices of car change.

4.1.2 Income Process

Households face an exogenous and stochastic income process. Household i at time t is at the age j_{it} has the income process as the following:

$$Y_{it} = exp\{\chi(j_{it}) + y_{it}\},$$

throughout the life-cycle, where $y_{i,t}$ is the residual income, and $\chi(j_{it})$ is the deterministic age-dependent parameter. z_{it} is household permanent shock that follows an $AR(1)$ process below:

$$y_{it} = z_{it} + \varepsilon_{it}$$

$$z_{it} = \rho z_{it-1} + \mu_{it}.$$

4.1.3 Per-period Budget Constraint and Adjustment Costs

The per-period budget constraint without adjustment costs is

$$C_{it} + P_t(D_{it} - (1 - \delta)D_{it-1}) + A_{it} = Y_{it} + (1 + r)A_{it-1}.$$

The adjustment costs incur only if households decide to adjust their durable level. Two types of adjustment costs are included: the proportional adjustment costs

(F_v) and fixed adjustment costs (F_o). The proportional adjustment cost, F_v, is the transaction cost that is proportional to the value of cars being sold. It captures the time and psychic costs of selling a car, as well as the possible loss in value due to asymmetric information. Note that the cost only depends on the size of existing durable, not the new purchase. This will give a policy rule that depends only on the state variable, and not on the choice variable. Moreover, an additional fixed cost, F_o, is included to capture the value-invariant costs of selling a car, such as title transfer and license registration fees.

Both types of adjustment costs are effective in creating infrequent adjustment of durable goods. However, the proportional adjustment cost will result in a policy function that is dependent on the size of state variable, D, while the fixed adjustment cost allows for varying frequency of adjustment among wealthy and poor households. The proportional adjustment cost enters the first order condition directly, and affects households with varying level of wealth by the same extent. On the other hand, the fixed adjustment costs do not enter the first order condition, but it has what is akin to a wealth effect. The fixed adjustment cost will create a varying degree of frequency of adjustment depending on household wealth level. Having two types of adjustment cost parameters will allow for flexibility in matching aggregated moments of frequency of adjustment among households with different levels of income.

The total adjustment cost given durable holding D_{t-1} is equal to

$$F = F_v P_t D_{t-1} + F_o$$

4.1.4 Terminal Value of Wealth

Townsend Thai data reveals that Thai households still hold a high level of assets even if they are near the end of life. The lack of asset deaccumulation indicates that households may have bequest motives, or in general they value having some assets at the end of their lives. In order to match this empirical observation, households are assigned a terminal wealth value.

For simplicity, I maintain the structure of the utility function and assigned a scaling parameter, ψ, as the terminal wealth value function:

$$V_T = \psi \frac{[(A_T)^{\alpha}(D_T + \tau)^{1-\alpha}]^{1-\sigma}}{1-\sigma}$$

4.1.5 Recursive Form

The household dynamic problem described above can be written in a recursive form as follows. Let the set of state variables $s \equiv (A, D, z, j)$. The state variables include household wealth, durable holding, permanent income shock and age, respectively. In each period, households face a decision of whether or not to adjust their durable stocks. The discrete decision is made by comparing the optimal value of each sub-problem. Households face different budget constraints depending on whether they adjust their durable holding. In each sub problem, they also face the liquidity constraints and nonzero consumption constraints.

The value function given state variables is

$$V_t(s) = \max\{V_t^{adjust}(s), V_t^{no\ adjust}(s)\}$$

.

Households solve the following dynamic problem if they adjust their durable holding:

For $t = 1, ..., 60$:

$$\begin{aligned}
V_t^{adjust}(s) &= \max_{C,A',D'} U(C, D') + \beta E[V_{t+1}(s')|z] \\
s.t. \quad A' + PD'(1+\tau_d) + C(1+\tau_c) &= (1+r)A + y(z)(1-\tau_y) + (1-F_v)(1-\delta)PD - F_o \\
A' &\geq \underline{A}_{t+1} \\
D' &\geq 0 \\
C &\geq 0 \\
s' &= (A', D', z', j+1).
\end{aligned}$$

Where τ_d, τ_c, and τ_y are the (effective) tax rates on cars, non-durable goods, and labor income. There are two choice variable, A' and D' if households adjust their durable consumption, and C is complying with the per-period budget constraint and the nonzero constraint. D' is nonzero but can be lower than the previous durable stock net depreciation ($D(1-\delta)$) if households are selling their durable goods as a buffer stock and adjusting their durable level downward when they experience negative shocks.

Households solve the following program if they do not adjust their durable holding:

$$\begin{aligned} V^{no\ adjust}(s) &= \max_{C,A'} U(C,D) + \beta E[V(s')|z] \\ s.t. \quad A' + C(1+\tau_c) &= (1+r)A + y(z)(1-\tau_y) - \delta PD \\ A' &\geq \underline{A}_{t+1} \\ C &\geq 0 \\ s' &= (A', (1-\delta)D, z', j+1). \end{aligned}$$

Households will make an adjustment decision by choosing the option with higher value function at given state variables.

4.2 Model Solution

The model is solved backwards from the terminal period. In each period, two problems are solved for the adjust and non-adjust cases. The value functions of the two problems are compared for given state variables and the solution with higher values is the solution for that set of state variables.

In the non-adjustment case, households continue to consume the existing levels of durables net depreciation from the previous period. A household essentially has one choice variable in this case, the non-durable consumption. The next period asset will be according to the budget constraint. At a given durable level, the next period level of asset or nondurable consumption is chosen to optimize the problem according to the intertemporal first order condition for interior solutions. Corner solutions may occur in the cases of either binding liquidity constraints, or zero consumption.

In the case of durable adjustments, households have two choice variables:

durable and non-durable consumption. The two choices variables will be determined by two first order conditions that dictate the intratemporal and intertemporal substitution. Similarly, corner solutions for either binding liquidity constraints and/or positive consumption levels are considered.

According to the utility function specification, households have $-inf$ utility if its non-durable consumption is zero. Besides the imposed borrowing constraints, $\underline{A}$, this feature of the utility function imposes a natural borrowing constraint on the asset level that households will avoid because they are heavily penalized when they cannot afford positive non-durable consumption.

The solution resembles the (S,s) bands optimal rule. To visualize the solution to this problem, it is useful to think of the solution to the model without adjustment cost as a benchmark. If households can adjust their stock of cars costlessly, they will re-optimize their durable level every period given the two first order conditions and constraints. For a given set of state variables there is an optimal level of durable goods, $D^*(A, D, j, z)$ and asset, $A'^*(A, D, j, z)$, that solve the problem V^{adjust} with no adjustment cost. The introduction of adjustment costs essentially creates an inaction band around this optimal point D^*. The larger the adjustment costs, the larger the band is going to be. At a given set of state variables, households will adjust when their durable levels become too low or too high outside of the inaction band.

Given a level of asset, a household will adjust durable upwards when the level of durable becomes too low, generally as durable depreciates over time. A household may also adjust its durable level downward in the case that its asset level become too low, possibly due to negative shocks. Durable goods essentially function like a buffer stock for the household. On the other hand, if there is a sufficiently large positive shock, a household will adjust its durable stocks upwards.

An example of a policy function given a state space of (A, D) is illustrated in Figure 9. The flat plane in the middle illustrates the area of (A, D) that households will not adjust their durable. In the right corner is where households will adjust the durable level upward, when a given level of D is too low for the level of asset. On the other hand, the left corner shows the area of the state space (A, D) that households

adjust their durable downwards. When their asset levels are too low, households resort to selling cars as if it is a buffer stock.

Figure 10 illustrates a level set of policy function of durable adjustment decision for 55-year-old households ($t = 30$), given that their permanent shock is 0 (so they are at a mean income). The level set shows the household policy function at different asset levels. First, if households have low assets and are very close to the liquidity constraints, $A = -198$, the policy function is to sell all the durable goods except for where their durable levels are extremely low and may not cover the fixed adjustment cost. If their durable holding is too high, the policy function is to liquidate existing durable goods and adjust the level of the durable holding downward. For a household with a low level of assets but sufficiently far away from the borrowing constraint, $A = 10$, the policy function is to not adjust and keep consuming the existing level of durable goods. As the level of wealth rises, the demand for durable goods rises. Households with higher levels of wealth will have larger areas of adjustment, that is the lower-bound threshold of durable in the inaction band will be higher as their wealth rises.

Figure 11 illustrates a level set of durable demand at different ages, given a level of assets at 200 and permanent shock $z = 0$. Demand for durables rises with age. Given the same level of asset, the desired durable holding is larger as households grow older due to diminishing precautionary saving incentives as they are closer to the end of the life cycle. The level of desired durables exceeds the actual asset holding due to the fact that households are allowed to borrow and durables are long lasting and provide a stream of services over time. Furthermore, in order to avoid frequent adjustment, households have incentive to purchase larger stocks of durables to avoid adjustment costs.

4.3 Computation

To compute the policy function and value function, I solved the model backwards from the terminal period. The state space for durable goods and assets is discretized to 30 grids each: $a_1, a_2, ..., a_{30} \times d_1, d_2, ...d_{30}$. The value of the maximum and minimum grids in each period of the life cycle is based on the range of asset and durable holdings at

each age from Townsend Thai data. The permanent shocks are also discretized into 9 states using the Tauchen method. The policy function is then evaluated at each state variable on the grids, although the solutions are not restricted to the same grids. Due to the highly discontinuous nature of the value function, I solve the dynamic problem by using a brute-force grid search. Details on the steps to solve the model are as follows.

1. The terminal period assets are constrained to be non-negative. The last period solution is solved given the terminal period function of wealth.

2. Given the state variables, the non-adjust problem in period 60 is solved by searching over next period asset values. First, the cash on hand available given that the household will not adjust durable level is computed. The cash on hand is then discretized into fine grids and the solutions are searched over the grids. Nondurable consumption is computed by per-period budget constraint and is constrained to nonzero value. The grid of next period asset level is constructed so that nondurable consumption is nonzero. If the implied consumption is negative given the cash on hand, a big negative value is assigned to the value function with a larger negative value when the asset is more negative. The solution is the grid that yields the highest value.

3. A similar procedure is followed for the adjust case. However, now there are two variables to search over: the levels of adjusted durable goods and assets. To solve for the adjust case the cash on hand given that existing durable goods are sold is calculated. The cash on hand then is discretized into fine grids to search over next period asset level. For each grid of the next period asset level, a set of fine grids of possible adjusted durables is constructed given that consumption level must be positive. The level of consumption is computed by the per-period budget constraint. The solution is the grid that yields the highest value.

4. For each set of state variables, the value functions between the two cases are compared to decide whether households will adjust the durable levels. Value function for $t = 60$ is stored.

5. Solve the model for $t = 59$, following the same procedure. The new set of grids of state variables are calculated according to data. The values of continued value

function over the fine grids that are searched are calculated using linear interpolation.

6. Repeat the process for $t = 58, ..., 1$ following the same procedure.

Extrapolation: To allow for solution in t to fall outside the maximum value of the already specified grids in $t+1$, extra grids with very large values of asset and durable levels are added and solved for to allow for the solution in $t+1$ to be extrapolated. Solutions that fall outside the grids specified for period $t+1$ are estimated by using linear interpolation to the value at the extra grids.

The algorithm is vectorized to maximize efficiency. The total time to solve the model for one set of parameters is approximately five minutes using Matlab. To estimate model parameters, the program is set up for parallel computing to search over parameters. The program uses Matlab software over supercomputing facility (HPC) provided by Research Computing, Arizona State University. It takes approximately 2 days to estimate one set of parameters.

4.4 Interpretation of Parameters

The standard Cobb-Douglas preference implies that the elasticity of intratemporal substitution (ES) between durables and non-durables is one, whereas the preference weight of nondurable goods is equal to α. Moreover, the curvature parameter σ is equivalent to the relative risk aversion, while the EIS is equal to the inverse of the relative risk aversion, $1/\sigma$. However, the addition of nonhomotheticity parameter in this model changes the standard interpretation and elasticities implied by the Cobb-Douglas utility function.

4.4.1 Non-homotheticity Parameter

The non-homotheticity preference introduces an income effect of cars, which would otherwise be absent in the Cobb-Douglas utility function. The parameter τ measures how much of a luxury goods a car is in household preference. The higher the τ is, the less of a

necessity cars are for households. The income effect is an important aspect of the model to fit a relative demand function for durable goods [Pakoš, 2011]. A consumption share of nondurable goods will be relatively large when the value of τ is large. At sufficiently low income (wealth), households will have zero demand for durable consumption. At this point, a small change in relative prices of durable goods will have no impact on non-durable demands, hence zero elasticity of intratemporal substitution. As household income (wealth) becomes sufficiently high, the ES is larger and approaches 1.

A low value of τ implies that the outside option of owning a private vehicle provides low utility flow or that there is not many non-durable goods or services that can be used in place of cars. Figure 12 demonstrates the aggregate implications of the parameter over the average life-cycle consumption and saving. At low value of τ, cars are more of a necessity and hence households will start purchasing a car very early in life. This contributes to a relatively low level of assets and non-durable consumption. The lower the τ, the higher the fraction of car owners in the population. They will also have higher cumulative durable adjustment and frequency of adjustment over the life cycle. For instance, if τ is only 50, all the median-income households will own a car. When the value of τ is high, young households are more likely to not own a car and become car owners when they are older and accumulate more assets.

Note that the model-generated fraction of the population owning a car is monotonically increasing over the life cycle given any value of τ due to the imposed utility function and the life-cycle setting. To allow for a more flexible pattern of the fraction of car owners generated by the model to match data, cohort-specific τ would be necessary. This is because the data consists of many cohorts, and different cohorts could have varying degrees of non-homotheticity in their car preferences.

4.4.2 Curvature Parameter and EIS

Given a standard constant relative risk aversion (CRRA) utility function, σ is the coefficient of relative risk aversion, and EIS is equal to $1/\sigma$. However, given the non-homotheticity in durable good preferences, σ is no longer equivalent to the coefficient of relative risk aversion, and $1/\sigma$ is no longer the EIS. Nonetheless, their inverse relation-

ship remains. For clarity, the parameter σ is referred to as the curvature parameter.

Households with high curvature parameters, σ, will prefer having smooth consumption over their life cycle. Generally, a high value in the curvature parameter works in two directions: 1) households have precautionary savings to smooth consumption over time, and 2) households whose EIS is low prefer a flat consumption profile. Consequently, they may consume more when they are younger in comparison to households with lower value in the curvature parameter. Households with low value of the curvature parameter or high EIS may hold off consumption when younger and consume a large amount later in life.

In the presence of durable goods, households are also likely to smooth out durable consumption over the life cycle. Therefore, households with high curvature, or low EIS, will be more likely to buy durable goods when they are younger in order to smooth out their durable consumption over the lifetime. This in turn will result in less saving, and lower consumption of both their durable and non-durable goods when they are older. On the other hand, households with low curvature parameter, or high EIS, will consume less durable goods when they are young, save more and enjoy a higher level of durable and non-durable goods when they are older.

At the aggregate level, the curvature parameter influences the trend and level of fractions of car owners, durable spending, and the frequency of durable adjustment in the economy over the life cycle. A higher level of the curvature parameter (lower EIS) will imply higher spending on both durable and non-durable goods at a younger age, and lower spending at an older age (Figure 13).

One concern is the the specified utility function has an underlying assumption that households have the same curvature in preferences for durable and non-durable goods. The functional form is in line with the standard in the literature; however, it imposes the same curvature on consumption profiles for both goods and generates the trend of life-cycle nondurable, durable, and assets mentioned above (Figure 13). Future work may consider relaxing this assumption to allow for the curvature of non-durable and durable profiles to vary more independently.

5 Empirical strategy

Some parameters are chosen outside the model from data or closely related studies. The income process parameters are estimated exogenously from the model. Given the previously chosen and estimated variables, the rest of the parameters are estimated endogenously within the model using GMM to match moments from data. Parameters chosen outside the model include r, δ, F_v, F_o, and β. The six parameters estimated within the model include $F_v, F_o, \tau, \alpha, \sigma$, and ψ.

5.1 Data

To estimate the model parameters, data moments are constructed from the household-level data, Townsend Thai Data: Urban and Rural Resurvey in 2005 to 2015. Townsend Thai data consists of a very detailed and continuous panel of household assets, consumption, and income for over 10 years. It is the longest and most complete panel of Thai household-level data. Moreover, assets and wealth data, which are the model's key variables, are of high quality and very detailed. The data are collected annually and if there is any discrepancy in questionnaire responses households will be asked to confirm or reconcile the discrepancy. This substantially reduces measurement errors normally encountered in the survey data.

One drawback is that the Townsend Thai data sample is not representative of the whole country while it is representative of the urban and rural areas of the six provinces it is sampled from. Nonetheless, summary statistics (Table 1) shows that the Urban and Rural Resurvey of Townsend Thai Data together are approximately representative of the Socioeconomic Survey (SES) data, which is the primary Thai household data collected by the National Statistical Office. For instance, the averages of gross income and the number of vehicles owned by households in SES fall right between the averages of those in Townsend Urban and Rural Resurvey.

5.1.1 Assets

Table 2 shows summary statistics and the composition of household assets. Household assets included as the variable A in the model are comprised of both liquid and illiquid assets. Liquid assets include saving, checking accounts and cash. Illiquid assets consists of household fixed assets, vehicles, land, land with housing, agricultural assets, business assets, lending, and net liability.

As mentioned before, the inclusion of illiquid assets is used to accurately represent household wealth. The average of liquid asset holdings for households is 27 thousand THB, which accounts for only 6% of the average household net wealth of 436 thousand THB (Table 2). Household asset holding would be tremendously misrepresented without including illiquid wealth. Given that the model period is one year, the assumption that households can access their illiquid wealth costlessly is reasonable. However, in a model concerned with shorter responses, such as quarterly responses, the assumption that these assets can be accessed without friction might be too strong.

A look at the average and median of the composition of household wealth reveals a large wealth inequality in Thailand. The average of wealth is significantly larger than the median wealth, and the majority of households do not own land, housing, or vehicles. The income disparity, however, is not as large. More than half of households do not own a car. The average of household vehicle value is 63 thousand THB, approximately 15% of the average net wealth (Table 2).

The life-cycle asset profile from Townsend data (constructed by substracting out the year effects) shows a similar hump shape observed in the US data. Wealth accumulation peaks at around age 50, but older households barely deaccumulate wealth afterwards (Figure 16).

<u>Fixed Assets and Personal Vehicles</u>

Townsend data reports initial asset value in the first survey, and all the new purchases of fixed assets in the following resurvey. To construct a household fixed asset profile, I use depreciation rates specific to types of fixed assets. Household board durables, including appliances, are assumed to depreciate at rate 16.5 %, agricultural

assets at 13.15%, and business assets at rate 15.79 %. The rates are chosen based on the depreciation rates used by the U.S. Bureau of Economic Analysis on the most related board durables in each category. Fixed asset value and household car value are calculated following the standard law of motion:

$$d_{it} = (1 - \delta_d)d_{it-1} + I_{it}$$

Where d_{it} is the fixed asset or car value in each period and I_{it} is the new purchase of the fixed asset at time t.

For the purpose of the car tax rebate policy evaluation, only passenger cars and pick-up trucks are included in the value of durable consumption, D_{it}, in the model. Other types of fixed assets are included in household wealth, A_{it}. The depreciation rate for cars is assumed to be 9.4%, from the author's calculation to fit the aggregate capital stock and gross capital investment of personal vehicles in the private sector as reported by the Capital Stock of Thailand and National Income (NESDB,2017).

Figure 17 shows household car holdings over the life cycle. The profile also shows a similar hump-shape pattern with a peak at around age 35. Afterwards, households seem to maintain a flat value of car holding over time. The frequency of car adjustment over the life cycle is also calculated by using the number of times that households purchase and/or sell a car over 10 years. It shows a declining trend that younger households tend to purchase a car more frequently than older households (Figure 19.) This may reflect cohort-specific preferences that cars are more of a necessity for younger cohorts.

5.1.2 Consumption

Household non-durable consumption consists of household spending on food, alcoholic beverages, tobacco, gasoline (excluding gasoline for business or farm use), and spending on ceremonies. Household spending on services and semi-durable goods, namely house repairs, vehicle repairs, education-related expenses, clothing, and food eaten away from home are also included. Figure 18 shows household non-durable spending by cohorts from 2005 to 2015. Non-durable consumption also has a hump-shape profile with a peak

at age 50.

5.1.3 Income

The labor income is constructed from wages and gross operating surpluses of households from the Townsend Thai data. For households that are wage earners, the labor income is straightforward. However, more than half of the sample are business owners and farmers. Their earned incomes, i.e. operating surplus, is a mixed income: a return of both labor and capital. In order to account for labor income for these households, I account their income based on the capital productivity factor reported in the Capital Stock of Thailand by the National Accounts Office. I use the capital product factor for the agricultural sector (labor product factor = 1-capital product factor) to account for operating surpluses of farms, and the capital product factor for the service sector for operating surpluses of other businesses. (Note that most small businesses that households own are in the service sector while most of workers in the industrial sector are wage earners.) If households have many sources of income I add all the incomes that are considered labor income plus a fraction of operating surpluses of their businesses or farms together. Income also has a hump-shape profile over the life-cycle (Figure 15), with a peak also at age 50 and a steady decline afterwards.

5.2 Estimations

Estimations are done in two steps: first some parameters are chosen or estimated outside the model. Then given the exogenously chosen or estimated parameters, the rest of parameters are estimated endogenously within the model. Parameters chosen outside the model include r, δ, F_v, F_o, and β, while the income process parameters are estimated exogenously. There are five parameters estimated within the model, including $\tau_{cohort=1,2}, \alpha, \sigma$, and ψ.

<u>Parameters Chosen Outside the Model</u>

The asset return rate, r, is chosen to reflect a middle range value of asset returns in the data. Assets in the model consist of both liquid assets, in the form of

saving and cash, and a household's fixed assets net liability. The average saving rate in Thailand over the time period was relatively stable, around 2-3 % annually (author's calculation from saving interest rates of major commercial banks reported by the Bank of Thailand). Some households own businesses or farms and business or farm assets often generate returns at a higher rate than saving. Furthermore, some households lend money to other households and their reported returns are approximately 12 % (weighted average by the size of loan). Given the return rates from various sources, I choose a middle value of asset return rates at 5 %.

The depreciation rate of cars is chosen to be at 9.4 %. The rate is calculated by the author given the aggregate level of capital for private vehicles and the level of the gross private personal vehicle investment reported by National Accounts over the past 10 years. The depreciate rate is calculated to match the implicit depreciation rate over time of the aggregate level of private personal vehicles.

The fixed adjustment cost F_o is set at 8 thousand THB, reflecting expenses in purchasing a vehicle, including title transfer, and license registration. The proportional adjustment cost, F_v, is chosen to be 0.10. However, robustness check is made for assumed values of F_v equal to 0.05 and 0.15.

β is fixed to be at 0.95. This is the value that yields the best fit of moments on average and lowest residuals of the quadratic of the differences between model and data moments.

Income Process Parameter Estimation

In order to fit the stochastic parameter for the income process, first the life cycle component is estimated to fit a 4th order polynomial regression of yearly earnings on age. The year effect is also estimated as a deterministic part of the income. The residues from the regression are then use to estimate the stochastic process parameters following the estimation process as in Kaplan and Violante [2010]. The parameters are estimated using GMM, by matching the covariance matrix of regression residues of yearly earning to the covariance matrix generated from the model. The permanent shocks are estimated to follow the AR1 process with an autocorrelation ρ_z=0.93, and a standard deviation of $sigma_z$ = 0.09, while the temporary shocks have a standard

deviation $\sigma_x = 0.29$ (Table 5.)

5.3 Identification

To pin down the key parameters of interest, namely the curvature and non-homothetic parameters, adjustment costs parameters, F_v and F_o, are chosen to fit outside evidence, and the patience parameter β is fixed at 0.95. This is because the consumption and asset profiles cannot pin down the adjustment cost, patience, curvature, and non-homothetic parameters at the same time. For instance, there are many ways that the model can generate a high rate of consumption of durable goods early in life: 1) value of F_o is low, 2) τ is low, or 3) σ is high (EIS is low) and households prefer having durables early in life. Similarly, there are a few ways to generate high assets at the end of life: 1) having high bequest motive, ψ, 2) high patience, β or 3) low σ (high EIS) so households save in order to have high consumption later.

Without fixing F_v, F_o and β, the key parameters cannot be pinned down. Despite the fact that the moments of frequency of adjustment and probability of adjustments are also included, adjustment cost parameters are difficult to pin down concurrently with τ because both parameters influence the frequency of adjustment. However, in future work that are interested in pinning down the dynamic of durable adjustment more precisely, the frequency and size of adjustment can help pin down these parameters while σ and τ are fixed or estimated exogenously with proper identification. An indirect inference that focuses on the dynamic of the durable adjustment, e.g. see Berger et al. [2018], is also another alternative to pin down the adjustment cost parameters.

The rest of the parameters are estimated using GMM to match model-generated moments to data moments. Targeted moments include 1) car ownership rates, 2) frequency of adjustment, 3) probability of adjustment, 4) asset, 5) non-durable spending and 6) durable spending. Durable spending is chosen over durable stock to be the matching moments due to the fact that durable stock is a model-generated value that is contingent on the assumed value of the depreciation rate, δ. Durable spending, on the other hand, is the value that is taken directly from data and is also more relevant for the policy evaluation of consumer spending.

Parameters F_o and τ are both influential to a household's decision of owning a car over the life cycle. F_o and τ will move together to match the level of durable spending. To pin down τ, I fixed F_o at 8 as a benchmark. Car ownership rates are another key moments that pin down τ, while the consumption and asset profiles together pin down the curvature parameter, σ, and the terminal wealth parameter, ψ.

Moments are targeted at $Age \times Income \times Year$ cells. Parameters are calibrated to match data in year 2006 to 2011, prior to the car tax rebate introduction. For robustness, data moments with sample size less than 30 are excluded, while moments of the probability of adjustment, durable spending and frequency of adjustment with sample sizes of car owners less than 10 are excluded. To avoid the small sample size bias, the identity matrix is used as the weight matrix for GMM. Moreover, due to the calculation of frequency of adjustment, a few periods of model needs to be simulated for the value of frequency of adjustment to converge. Therefore, only the last three periods of the moments of frequency of adjustment from the models are matched with data.

As mentioned above, the implied monotonically increasing trend in the fraction of car owners over the life cycle will impose additional restrictions on model-generated moments. Assuming one value of τ for all cohorts imposes an additional assumption that populations over different cohorts must have the same taste or preference for cars. However, the cohort effect is likely present in the data where young households are more likely to own a car due to their lifestyle than older households, despite having relatively low wealth or income. To capture the cohort effects, I allow the non-homotheticity parameters to vary by cohort: τ_1 for the young and middle-age cohorts, and τ_2 for the old cohort.

6 Results

6.1 Model Fit

Estimation results are reported in Table 5. As expected the non-homotheticity parameters are lower for younger cohorts than the old cohorts, in line with the fact that car usages are more prevalent among the younger households. The parameter α, which is the

share of nondurable consumption in a standard Cobb-Douglas utility function, is equal to 0.26. The value is smaller than the share of nondurable consumption in the data due to the non-homotheticity specification. The parameter α moves with the parameter τ to scale the size of non-durable consumption to the non-homotheticity parameter. The curvature parameter σ is estimated to be 0.95 and is robust to various specifications of F_v (Table 6.)

Figures 21 and 22 compare model-generated moments to data moments. Both the asset level and car ownership rates are matched fairly well throughout the life cycle for all levels of income (Figure 21) after allowing for the inclusion of the terminal value parameter and cohort-specific non-homotheticiy parameters. The model-generated moments of durable spending and non-durable spending also fit data moments reasonably. The level of durable spending on average is in line with data but the timing of durable spending is not exact. The probability of adjustment is matched well for most groups except for the young cohorts for whom the model underestimates their probability to adjust.

The model underpredicts durable spending and probability of durable adjustment among the young cohort, especially the low income group. Allowing for an additional young-cohort specific τ may improve the fit.

6.2 Elasticity of Intertemporal Substitution

In a standard Cobb-Douglas utility function, EIS is equivalent to $1/\sigma$. The estimated σ given other parameters is around 0.95, which would yield EIS of 1.05 with the absence of non-homotheticity parameters. Closely-related studies [Berger et al., 2018, Fernandez-Villaverde and Krueger, 2011] assume a middle range of commonly used value of σ to be 2, which implies a moderate size of EIS at 0.5. However, in a recent notable study Kaplan and Violante [2014] has chosen EIS (not estimated) value of 2 and argued that high EIS is consistent with theoretical and empirical evidence in recent promising approaches in the asset pricing literature [Bansal et al., 2012, Barro, 2009], and GMM-based methods that have estimate EIS well above 1 [Hansen et al., 2007].

Due to the income effects generated by the non-homothetic preferences, the

ES and EIS for households are different at varying level of income and wealth. At a low level of wealth or income, durables are a luxury and can be perfectly substituted by nondurables. Households must already be consuming a sufficiently high level of nondurables in order to start consuming durables. Then ES also influences the size of household EIS.

Besides the varying ES, adjustment costs in durables and liquidity constraints also create additional frictions in a household's durable adjustment. Frictions should not interfere with household long-run consumption; however, these frictions are important for shorter run responses within a few years of price changes.

EIS calculated from simulations is well above 1. Due to the rich features of the model, the closed-form solution of EIS is not possible [Attanasio and Weber, 2010]. To illustrate heterogeneity in EIS generated from the model, I conduct simulations of optimal consumption paths implied by the model when expected and unexpected changes in the real interest rate are introduced. The simulation is initialized with the same initial distribution explained in the Policy Experiments Section. Figure 25 shows an average consumption and saving path from the simulations when a permanent change of 0.5% increase in real interest rate is introduced. The paths compare the average consumption and saving paths when there is no rate change to the paths with expected and unexpected rate changes.

Households respond to the rate change differently depending on whether the rate change anticipated or unanticipated. When the rate change is anticipated, households lower both non-durable and durable consumption many periods before the actual rate change in order to accumulate assets in preparation for the rate change. After the rate change occurred, household increase durable spending while non-durable consumption remained lower than the baseline.

When the rate change is unanticipated households are predicted to reduce non-durable consumption and increase saving immediately after the change takes place. However, the effect on durable spending is delayed. The simulation shows no change in durable spending in the first year of the unexpected rate change possibly due to the frictions from adjustment costs and liquidity constraints. However, the drop in durable

spending is prominent one year following the unexpected rate change and is predicted to resume to a level higher than the baseline afterwards due to the rise in wealth from the rate hike (Figure 25.)

To illustrate the heterogeneity in EIS for varying ages and income levels, the change in consumption following the change in real interest rate is shown in Table 10. The simulation result suggests that low-income and younger households have higher EIS than high income and old households. They have the highest drop in consumption change after the rate increase. This is contributed to by the fact that the low-income households are closer to the liquidity constraints and that young households have stronger precautionary saving motives. Young households are also the group with the highest decline in durable spending in one year following the rate change.

On the other hand, old households in the top quintiles (3rd and 4th), as well as young and middle-age households in the 5th quintile of are found to have positive EIS, that is they increase non-durable spending when interest rate increases. This is likely because the wealth effect from the rate increase is stronger than marginal utility of additional saving. In other words, at the high level of wealth, marginal utility in the value function from additional saving is smaller than the gain from marginal utility from current consumption. Simulation results also suggest that overall EIS can be more negative in the second year after the unexpected change, due to the frictions from adjustment costs and liquidity constraints (Table 10).

7 Policy Experiments

This section presents results of three policy experiments: consumption tax break, durable tax break, and income tax break. First, a policy experiment to mimic Thailand's car tax rebate policy is conducted to evaluate the model prediction power. The simulation is conducted based on model parameters estimated targeting household consumption only prior to the car tax policy. Then this section proceeds with alternative policy experiments, namely a tax break on non-durable goods and income.

The economy is simulated from initial values of the household data in 2005.

The simulated economy consists of 100,000 individuals with initial values of income, assets, and personal vehicle holdings in 2005 from Townsend Thai data. The initial values are initialized for households age 26, 46 and 66 to represent the young (26 to 45), middle age (46 to 65), and old (66 to 85) cohorts respectively (Table 7.) Initial values of assets and incomes are drawn from the 10th, 30th, 50th, 70th and 90th percentiles in the data. The number of households are weighted according to the Thai population distribution in 2005 reported by Official Statistics Registration Systems, with 52%, 36% and 11% in the young, middle-age and old cohorts respectively.

All the tax breaks in the simulations are introduced at $t = 7$, or equivalent to year 2011. The age-dependent deterministic components are the same as those estimated from Townsend Thai data. However, to simulate the aggregate economy more closely, the income year effect is according to the trend of aggregate Compensation of Employees: the aggregated pure labor income reported in National Income (NESDB, 2017). For simplicity I assume a uniform tax rates of 25%, 7%, and 5% for durable goods, non-durable goods, and income respectively across the economy.

7.1 The Car Tax Rebate Policy Simulation

The car tax rebate policy is a temporary and favorable shock in the prices of durables. To implement the policy experiment in the simulation, the tax break is introduced unexpectedly and temporarily for two years. The simulation also mimics the policy regulation by imposing an additional proportional fixed cost of 20% of the purchasing price for five years after purchasing if household purchased a car during the policy.

Furthermore, to mimic the policy that tax rebates are only eligible for new car purchases, I imposed a rule to partially limit household purchases that benefit from the tax break for purchases with a minimum value of 500 thousand THB during the policy. This works well in mimicking the policy since only new car purchases benefit from the tax break. The limitations of this policy experiment and the model is that it does not capture the general equilibrium effect on car prices as a result of the policy. As discussed in Chapter 1, used car prices plummeted by 20% immediately after the policy had ended and prices of new cars also started to decline a few years after the policy had ended.

The result of this policy experiment should be interpreted as a pure partial equilibrium effect that the prices of cars are unaffected by the tax break.

The model is initialized in 2005. Between year 2005 and 2009, Thai income notably rose and the model predicted durable spending to increase dramatically (Figure 26), consistent with the observed distribution in 2009 in Townsend data (Table 8). The tax break is imposed unexpectedly in 2011 and 2012, and the car tax is resumed afterwards.

The model predicts durable spending to increase by a total of approximately 150% during policy. The result is consistent with the surge of new car sales during the policy shown in Chapter 1. However, the comparison of model durable spending with new car sales and national account data should be made with caution. New car sales and national account data only take into account household spending on new car purchases but not spending in the secondary market. However, the model durable spending includes both spending on new and used cars. The simulation result also suggests that the policy does not seem to cannibalize future household spending on cars by much, only 6% lower than the baseline after the policy. Yet this results do not contradict with the empirical evidence of cannibalization of future car sales presented in chapter 1, since the model abstract from the general equilibrium effect and the model durable spending also include spending on used cars.

The simulated data shows that most likely households to purchase a car during the policy experiment are those that already own a car with a high level of assets who make a large purchase during the policy. Some households with medium-level income and wealth also make a car purchase during the policy but they account for a smaller portion. Lower income and poor households do not make a purchase during the policy. This is expected due to the non-homothetic preference that most households purchasing cars during the policy are those with higher assets or income.

As expected, the policy has unintended effects on dampening nondurable consumption. The simulation predicts a sizable reduction in nondurable consumption during the policy of 5% in the first year, and approximately 1.5% lower in the following two years, and 0.8% and 0.2% lower in the fourth and fifth year (Table 11). However, larger

and long lasting impacts are on the decline in household assets. The simulation suggests as much as 7-8% decline in asset accumulation even after 2-3 years following the policy. The model assets include household fixed assets as well as household business and farm assets. The results strongly suggest that this could have a larger negative impact on the economy by dampening household investment and essentially cannibalizing the production growth and demand in other sectors. A general equilibrium model would be more apt to answer that question, however, the lower level of assets would also likely lead to lower investment and wealth in the general equilibrium model.

In conclusion, the model predicts that the car tax break contributes to additional car purchases of approximately 60-80% annually. The adverse effects on non-durable spending and asset levels are sizable. Future asset levels are predicted to be as much as 8% lower in at least the two years after the policy has ended. The sizable impact is contributed by to the presence of large ES among wealthy and high-income households and large EIS.

7.2 Alternative Policy Experiments

Similar policy experiments are conducted for non-durable tax break and income tax break. The tax breaks are introduced unexpectedly and lasted for 2 years.

Figure 27 shows the aggregate responses to the non-durable consumption tax break. The unexpected and temporary tax break of consumption tax of 7% is predicted to lead to approximately 8% increase in non-durable consumption during the policy, and a small increase of approximately 1% in the following 3 years after the tax break ended (Table 11.) Asset levels are predicted to increase as much as 8-9% during the policy and remained about 5-7% higher in the following 3 years after the tax break ended. There is a slight negative impact on durable spending, approximately 1% lower during and 2 years after the policy. However, durable spending with the consumption tax break is predicted to outgrow durable spending without the tax break in the third year following the policy.

Another policy experiment is the income tax break for two years (Figure 28.) The effective tax rate is assumed to be 5%. Given the progressive tax rates, in reality

only those that earn more than 200 thousand THB would have to pay income tax in Thailand. Given the income distribution, roughly only the top one quintile would have to pay income tax. Therefore, the assumption of 5% income tax rate is quite large for most people. The income tax break is predicted to increase both non-durable and durable spending by 2-4% during the tax break and the increase in consumption persists in the following years. Meanwhile, the assets are predicted to be as much as 10% higher (Table 11.) The surge in assets after the income tax break suggests that low income, low-asset households are likely to spend the additional income from the tax break on improving their asset balance sheets.

8 Conclusions

This study investigates the role of durable goods with adjustment costs in household consumption and saving decisions . Estimations shows a striking quantitative result that Thai households have high EIS and hence have large responses to the car tax rebate program. Due to large EIS, the stimulus has a large impact on shifting household future durable demand forward and significantly lower household future nondurable consumption and assets. Alternative policy experiments show that alternative form of tax break, such as consumption or income tax break will have a much more favorable impact on future consumption and saving.

This study also contributes to a few strands of literature, in particular, it quantitatively demonstrates the implication of non-homotheticiy preference of durable goods and transaction costs on EIS. Estimated EIS for Thai households is large and well above 1. Furthermore, contrary to previous empirical findings that wealthy households have higher EIS, the simulation results show that low asset and low income households could have large responses to a change in the rate of asset returns as well.

9 Tables

Year/Data set	**SES**	**Townsend Urban**	**Townsend Rural**
Variables	2009	2010	2010
Male	0.67	0.56	0.64
	(0.47)	(0.50)	(0.48)
Age	51.7	54.87	55.99
	(14.65)	(11.39)	(12.40)
College	0.11	0.17	0.04
	(0.32)	(0.38)	(0.19)
Gross Income (THB)	250,832	286,993	194,827
	(427,440)	(323,263)	(247,116)
Number of household members	NA	4.08	3.89
		(1.91)	(1.78)
Saving	NA	45,031	36,362
		(280,436)	(154,095)
Number of passenger cars owned	0.12	0.17	0.06
	(0.39)	(0.43)	(0.26)
Number of pick-up trucks owned	0.25	0.25	0.24
	(0.51)	(0.48)	(0.48)
Number of cars and pick-up trucks owned	0.37	0.42	0.30
	(0.65)	(0.64)	(0.57)

Standard errors in parenthesis

**Weighted average and standard errors adjusted so that the SES is representative to the kingdom*

Source: SES and Townsend Thai Data Annual Resurvey

Table 1: Summary Statistics of Thai Household Demographics in 2009-2010

	Mean (2002THB)	Mean Fraction of wealth	Median (2002THB)	Median Fraction of wealth
Income	163,153	0.37	100,640	0.92
Net wealth	436,012		109,195	
Net Liquid asset	27,322	0.06	3,660	0.03
Illiquid assets net liability	408,690	0.94	95,952	0.88
Household fixed asset	112,200	0.26	27,905	0.26
Vehicles	63,650	0.15	0	0.00
Land	400,974	0.92	0	0.00
Land with housing	42,990	0.10	0	0.00
Agricultural assets	11,096	0.03	0	0.00
Business assets	15,824	0.04	0	0.00
Borrowing	134,499	0.31	39,524	0.36
Lending	3,095	0.00	0	0.01

Source: Townsend Thai Data Annual Resurvey, 2005

Table 2: Thai Household Asset Portfolio Composition in 2005

	Young (25-40)	Middle Age (41-60)	Old 61+
Low Income	0.13	0.16	0.10
(1-3rd quintiles)	(67)	(475)	(435)
High Income	0.44	0.49	0.53
(4-5th quintiles)	(67)	(475)	(435)

Number of observations in parenthesis

Source: Townsend Thai Data Annual Resurvey, 2010

Table 3: Thailand's Car Ownership Rates in 2010 by Cohort and Income

	Young (25-40)	Middle Age (41-60)	Old 61+
Low Income	241	324	301
(1-3rd quintiles)	(67)	(475)	(435)
High Income	410	530	738
(4-5th quintiles)	(41)	(405)	(196)

Number of observations in parenthesis

Value in real term unit in year 2002 thousand THB

Source: Townsend Thai Data Annual Resurvey, 2010

Table 4: Thailand's Household Assets in 2010 by Cohort and Income

Chosen to match external evidence:	
r	0.05
δ	0.094
F_v	0.10
F_o	8
β	0.95
Income Process Parameters: Estimated exogenously	
ρ_z	0.9316
σ_z	0.0959
σ_x	0.2962
Estimated to match targeted moments	
τ_1	589.76
τ_2	652.32
α	0.2654
σ	0.9453
ψ	2.3402

Table 5: Parameter Values

Assumed F_v	τ_1	τ_2	α	σ	ψ
0.05	383.62	386.29	0.3673	0.9592	3.9421
0.10	589.75	652.32	0.2654	0.9453	2.3402
0.15	538.01	558.88	0.2763	0.9559	3.0323

Table 6: Robustness Check: Parameter Estimates with Varying F_v

Percentile	Young	Middle Age	Old
Income			
10th	20.3742	21.2512	12.5640
30th	45.9021	49.9741	34.8406
50th	79.9840	82.0556	60.3843
70th	117.2353	135.2242	113.9034
90th	216.1224	252.5160	239.1180
Asset			
10th	-83.2559	-100.6770	-57.9023
30th	-25.3541	-31.0156	-4.9154
50th	-5.5769	-1.7940	23.0558
70th	27.6121	65.4760	151.1958
90th	410.2276	475.8174	798.6896
Personal Vehicles			
10th	0	0	0
30th	0	0	0
50th	0	0	0
70th	0	0	0
90th	197.6870	256.1757	198.6870

Value in real term in year 2002 thousand THB

Source: Townsend Thai Data Annual Resurvey, 2005

Table 7: Distribution of Thai Household Income and Assets in 2005

Percentile	Young	Middle Age	Old
Income			
25th percentile	58.99	57.13	34.32
50th percentile	97.87	104.88	64.72
75th percentile	149.70	183.74	130.54
95th percentile	309.36	374.20	321.29
Asset			
25th percentile	6.61	16.89	21.82
50th percentile	123.96	185.56	142.61
75th percentile	426.49	508.06	479.03
95th percentile	1226.08	1368.98	1616.31
Personal Vehicles			
25th percentile	0	0	0
50th percentile	0	0	0
75th percentile	56.55	71.71	0
95th percentile	313.69	379.56	299.60

Value in real term in year 2002 thousand THB

Source: Townsend Thai Data Annual Resurvey, 2009

Table 8: Distribution of Thai Household Income and Assets in 2009

Years after the rate change	Unexpected change (1)	No change (2)	(1)-(2)	**EIS**
t=0				
Nondurable+Durable	-0.069	-0.0319	-0.038	**-3.98**
Nondurable	0.0142	0.035	-0.0231	**-3.98**
Durable	-0.167	-0.1677	0	**0**
t=1				
Non-durable+Durable	-0.318	-0.096	-0.222	**-46.71**
Non-durable	0.0521	0.0687	-0.0166	**-3.489**
Durable	-0.507	-1.762	-1.594	**-264.04**

Consumption growth rate is calculated from $\ln C_{it}/C_{it-1}$

Aggregate EIS is calculated from $\ln(\frac{\sum C_{it}^{r=0.055}}{\sum C_{it}^{r=0.05}})/\ln(\frac{1+0.055}{1+0.05})$

Note that changes in durable shown here only reflect durable spending only. If household adjust their durable stock downward, this is not reflected in the EIS of durable calculated.

Table 9: Aggregate consumption growth given an unexpected and permanent change of 0.5% in real interest rate.

Income quintile	1	2	3	4	5
Years after the rate change					
t=0	*Non-durable consumption*				
Young	-15.3838	-10.7006	-7.4673	-3.9251	0.0755
Middle Age	-9.6155	-7.5859	-4.4405	-4.4971	2.8819
Old	-2.1101	-2.0258	1.4505	3.7915	-2.1139
t=1					
	Non-durable consumption				
Young	-11.9989	-9.7911	-6.3606	-2.4030	-1.3866
Middle Age	-8.5881	-6.7969	-3.2335	-1.7099	0.6227
Old	-1.1840	-1.0907	1.0421	4.3409	-0.7523
	Durable spending				
Young	NA	NA	-332.5354	-246.0743	-388.0935
Middle Age	NA	-90.4792	-179.2275	-177.5845	-222.9534
Old	NA	NA	-163.0574	-88.2789	-61.3172
	Non-durable + Durable spending				
Young	-23.4761	-19.2970	-29.5814	-50.7130	-104.6438
Middle Age	-19.3670	-13.6164	-36.6854	-14.9088	-70.4760
Old	-2.8300	-1.3115	1.6549	-21.9404	-1.4356

EIS is calculated from $\ln(\frac{\sum C_{it}^{r=0.055}}{\sum C_{it}^{r=0.05}})/\ln(\frac{1+0.055}{1+0.05})$

Table 10: Heterogeneity in Consumption Responses from a Simulation Given Unexpected Real Interest Rate Change of 0.5%

Years after the policy introduction	0	1	2	3	4
Car Tax Break					
Non-durable spending	-0.0573	-0.0135	-0.0149	-0.0076	-0.0019
Durable spending	0.5794	0.8817	0.2308	-0.0646	-0.0592
Assets	0.0329	-0.0691	-0.0786	-0.0767	-0.0747
Nondurable Consumption Tax Break					
Non-durable spending	0.0853	0.0821	0.0118	0.0099	0.0088
Durable spending	0.0140	-0.0139	-0.0115	-0.0033	0.0205
Assets	0.0869	0.0788	0.0699	0.0622	0.0557
Income Tax Break					
Non-durable spending	0.0228	0.0207	0.0179	0.0151	0.0135
Durable spending	0.0385	0.0232	-0.0082	0.0575	0.0281
Assets	0.1007	0.1044	0.0920	0.0801	0.0712

Changes reported are calculated from $\ln(\sum X_{it}^{taxbreak} / \sum X_{it}^{notaxbreak})$

Table 11: Policy Experiments: Comparison of Aggregated Optimal Consumption and Assets

10 Figures

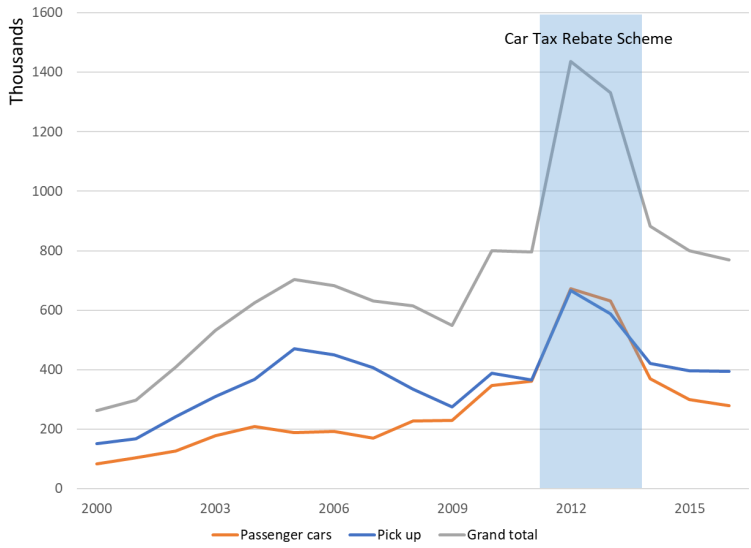

Figure 1: Annual new car sales in Thailand in 2000-2015

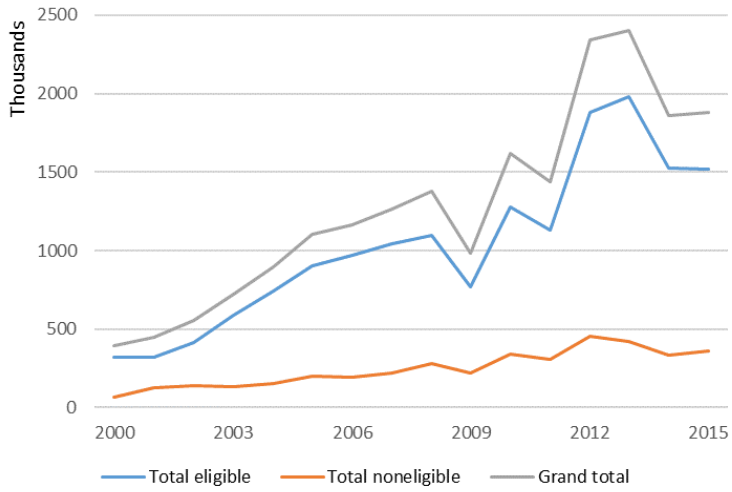

Figure 2: Domestic car production by eligibility to the car tax rebate in Thailand

Source: The Thai Automotive Industry Association (TAIA)

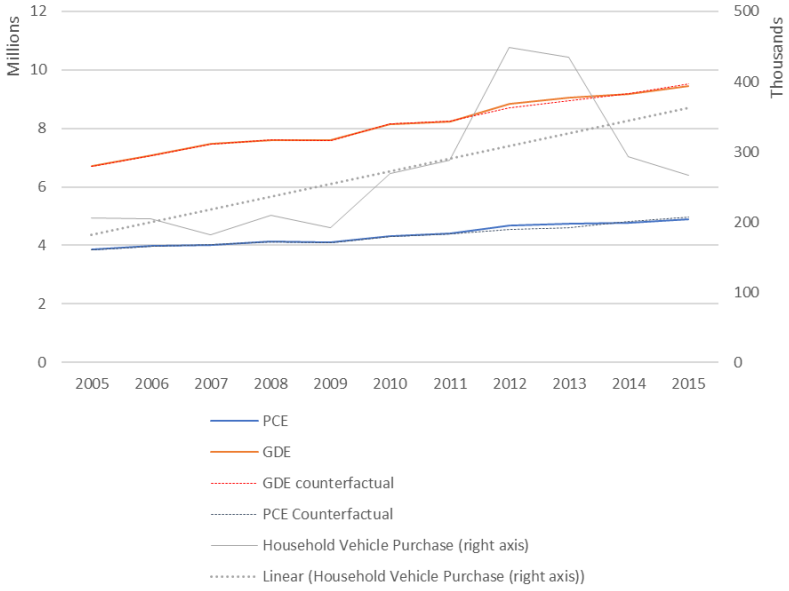

Figure 3: Thailand's GDE, PCE and Personal Vehicle Purchases in 2005-2015

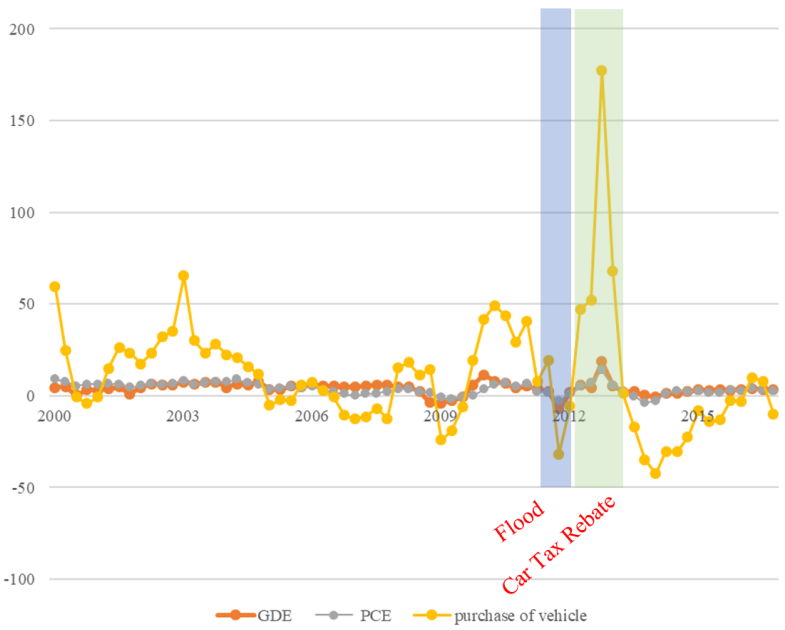

Figure 4: Growth trend of GDE, PCE and Personal Vehicle Purchase

Source: Office of the National Economic and Social Development Board (NESDB)

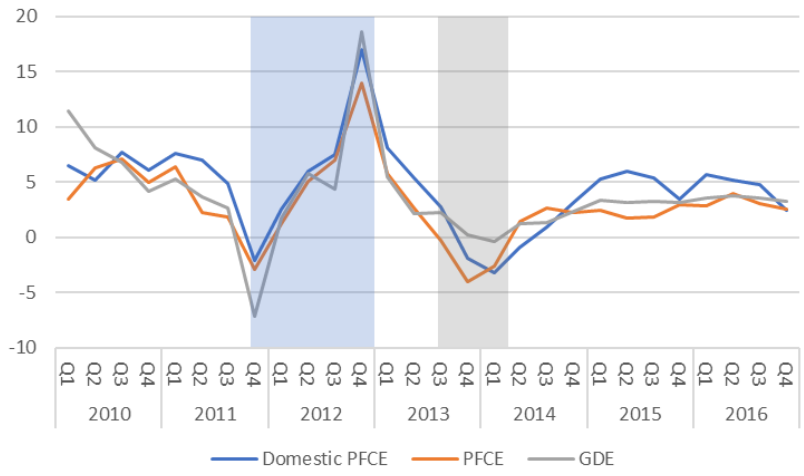

Figure 5: Growth of Thailand's GDE and PCE: a closer look

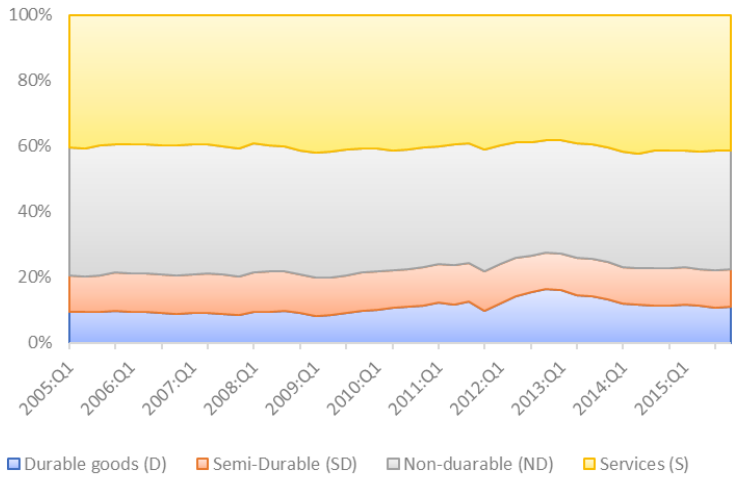

Figure 6: Thailand's Domestic Private Consumption Expenditure Components by Durability

Source: Office of the National Economic and Social Development Board (NESDB)

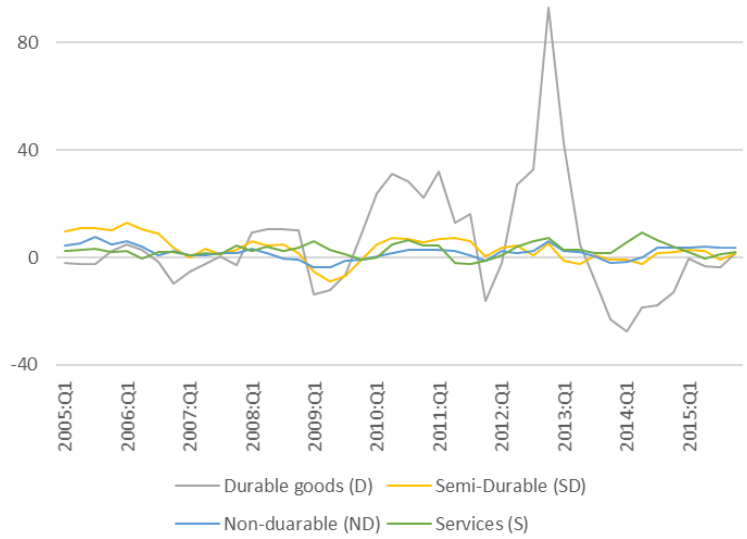

Figure 7: Domestic Growth of Private Consumption Expenditure Components by Durability

Source: Office of the National Economic and Social Development Board (NESDB)

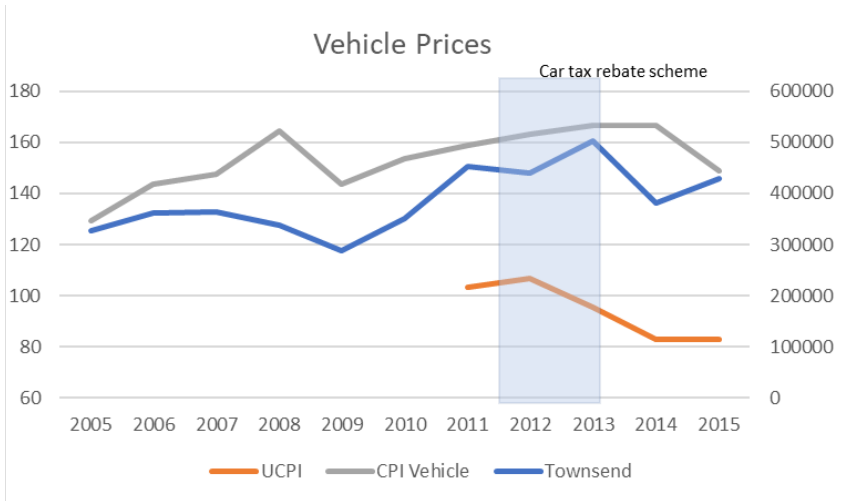

Figure 8: Personal vehicle price indexes

Source: The Bank of Thailand; Bureau of Trade and Economic Indices; Author's calculation from Townsend's Thai Data: Urban and Rural Resurvey

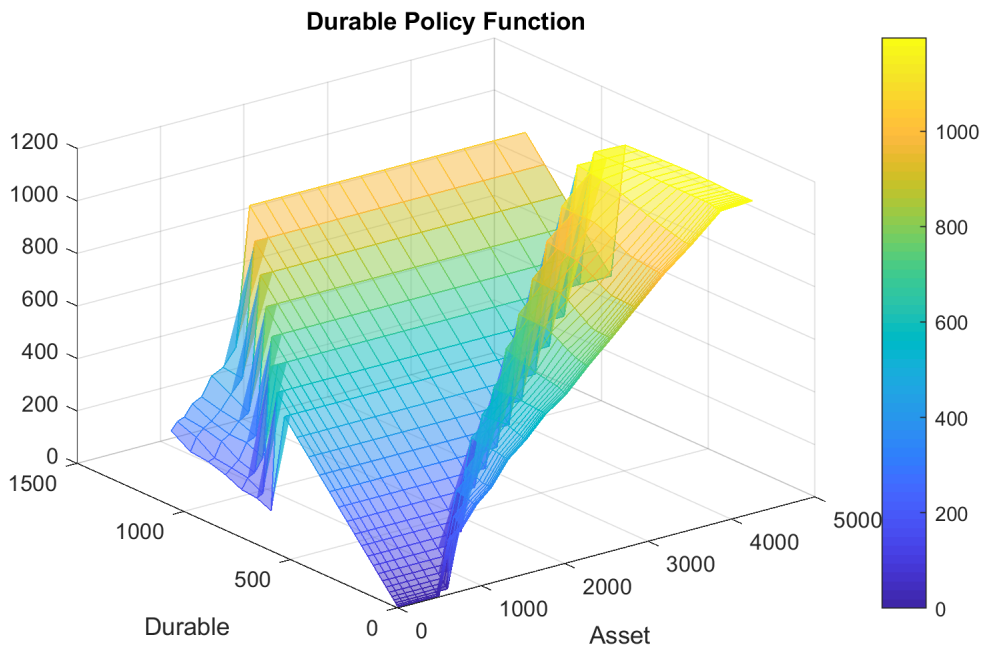

Figure 9: Policy Function of Durable Holding Given A and D

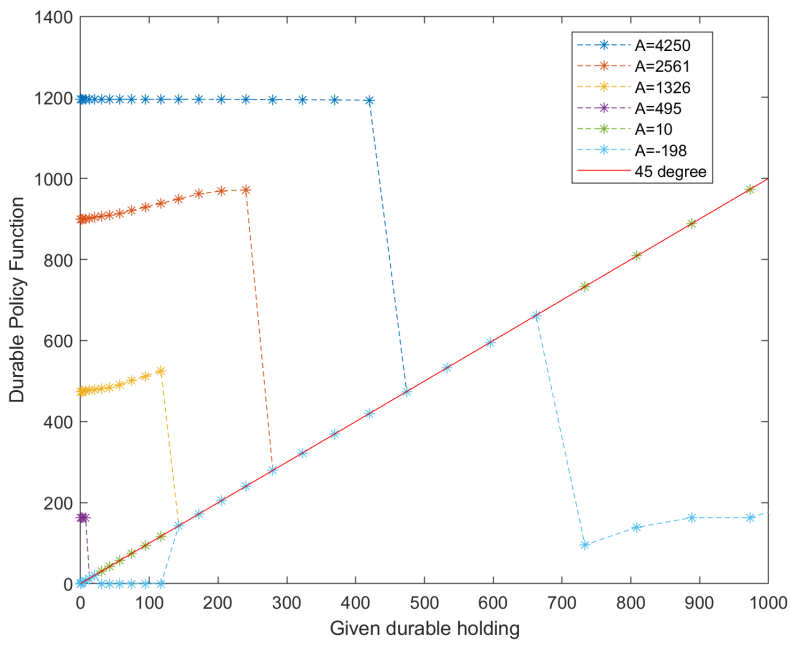

Figure 10: A Level Set of Policy Function at Different Level of Assets Given $t = 30$ and $z = 0$

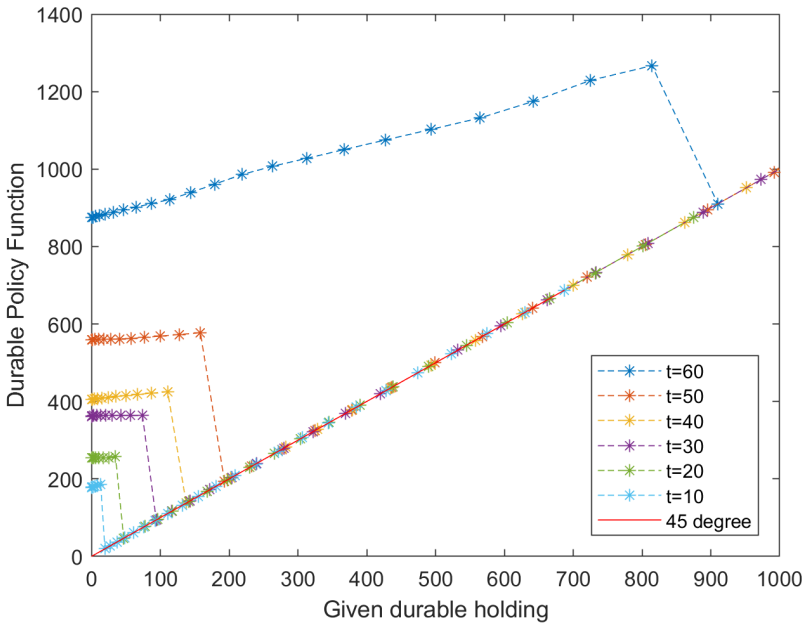

Figure 11: A Level Set of Policy Function at Different Ages Given $A = 200$ and $z = 0$

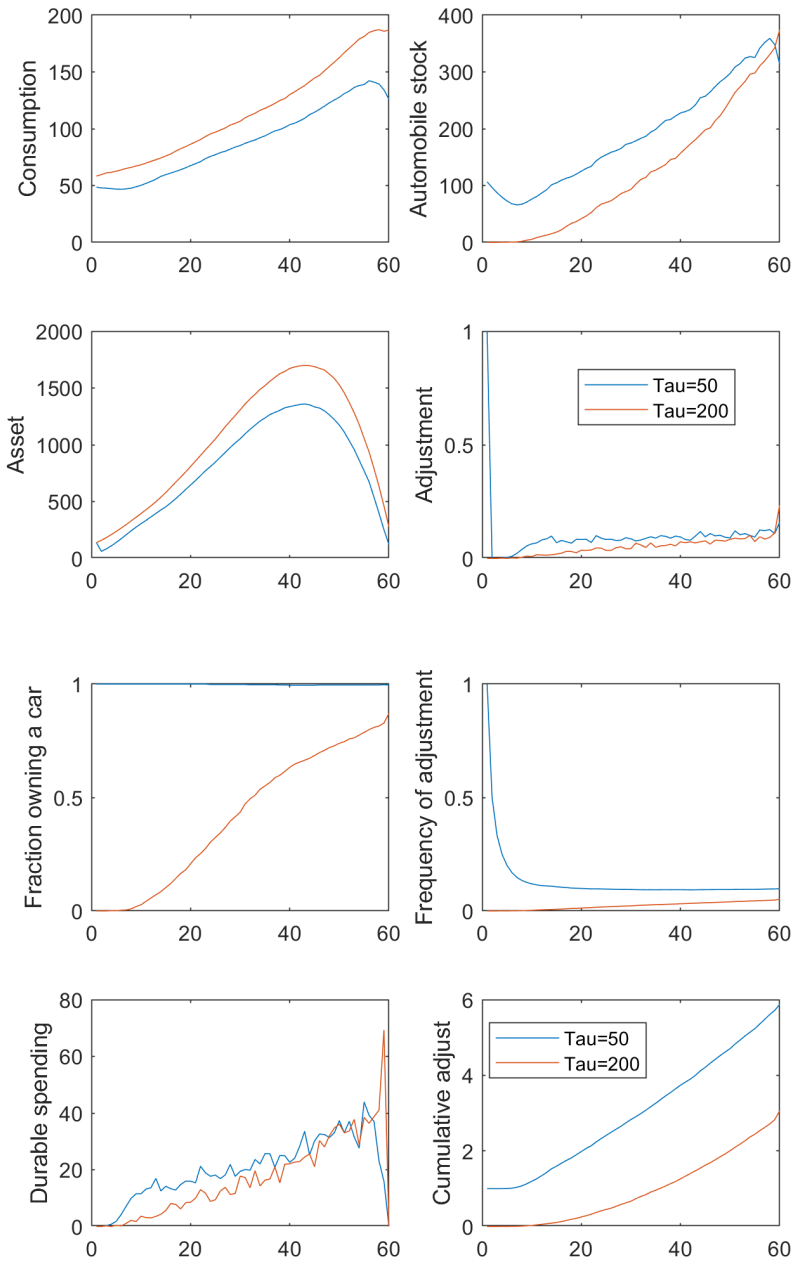

Figure 12: Comparative Statics for τ: Average Life-Cycle Consumption, Assets, and Durable Adjustment

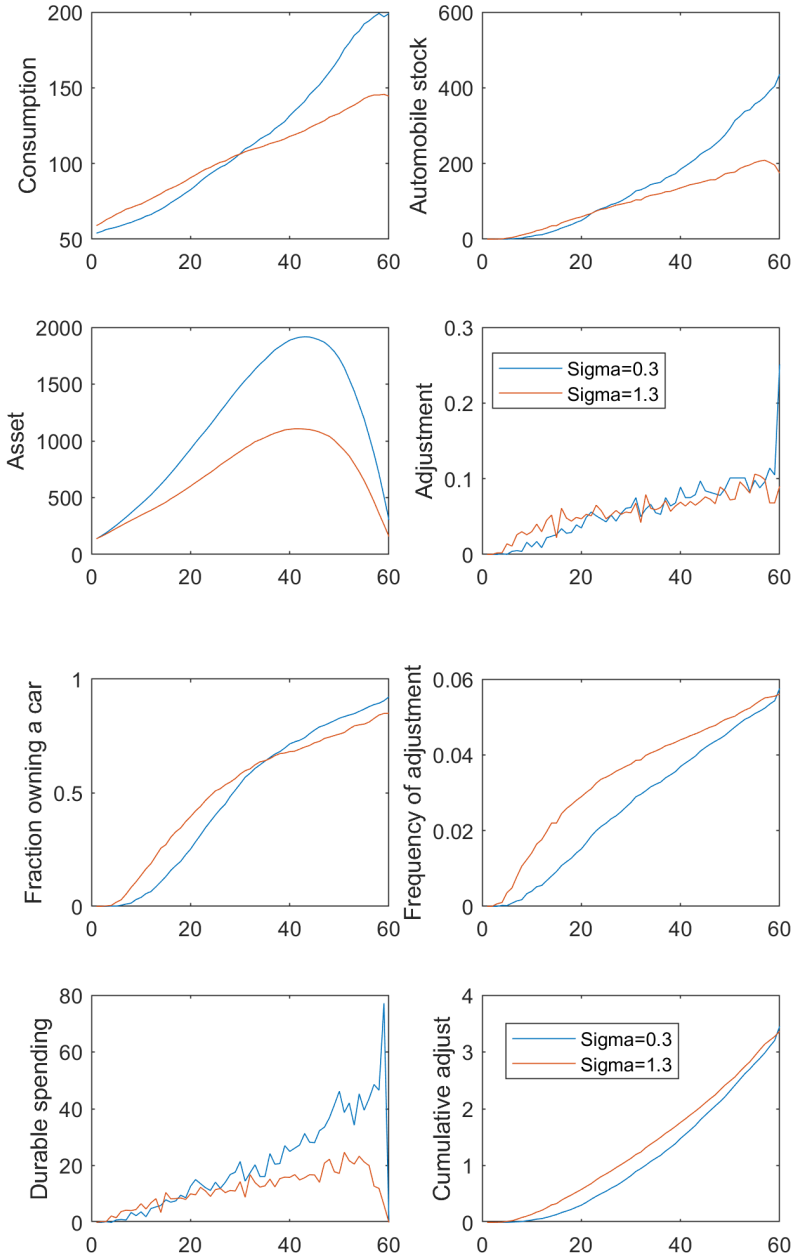

Figure 13: Comparative Statics for σ: Average Life-Cycle Consumption, Assets, and Durable Adjustment

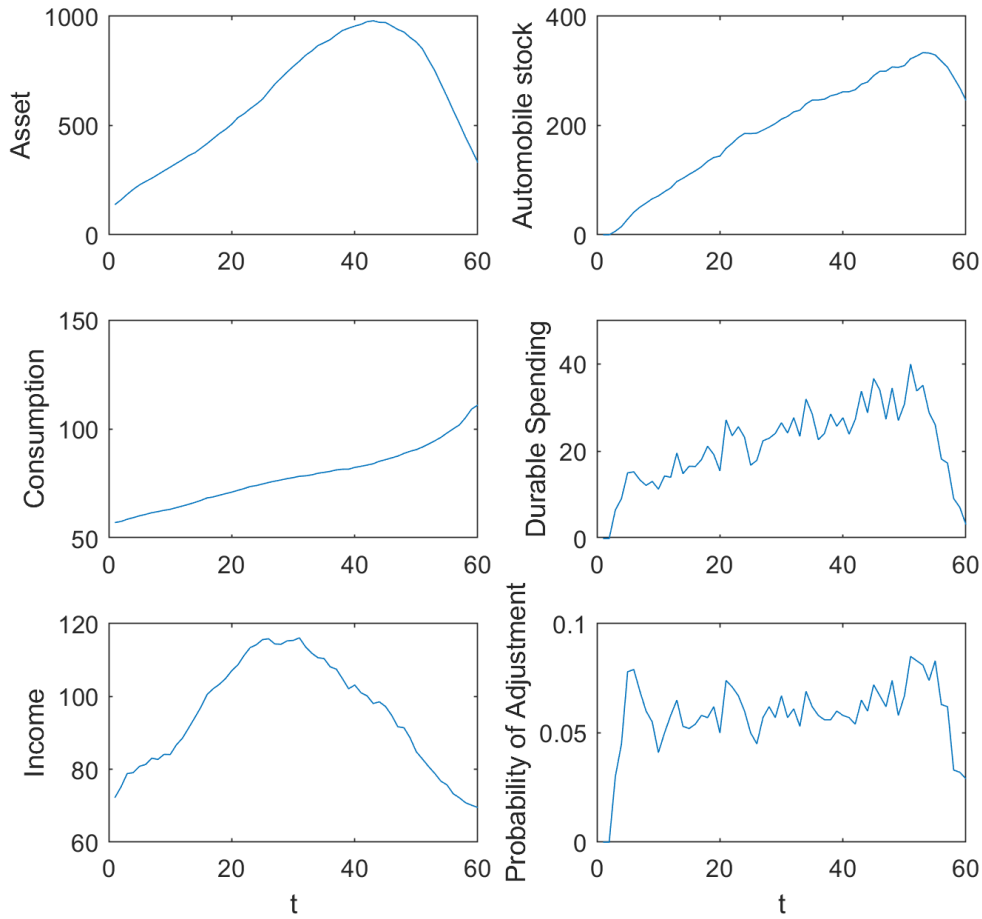

Figure 14: Average Life-Cycle Path Way of Median-Income Households

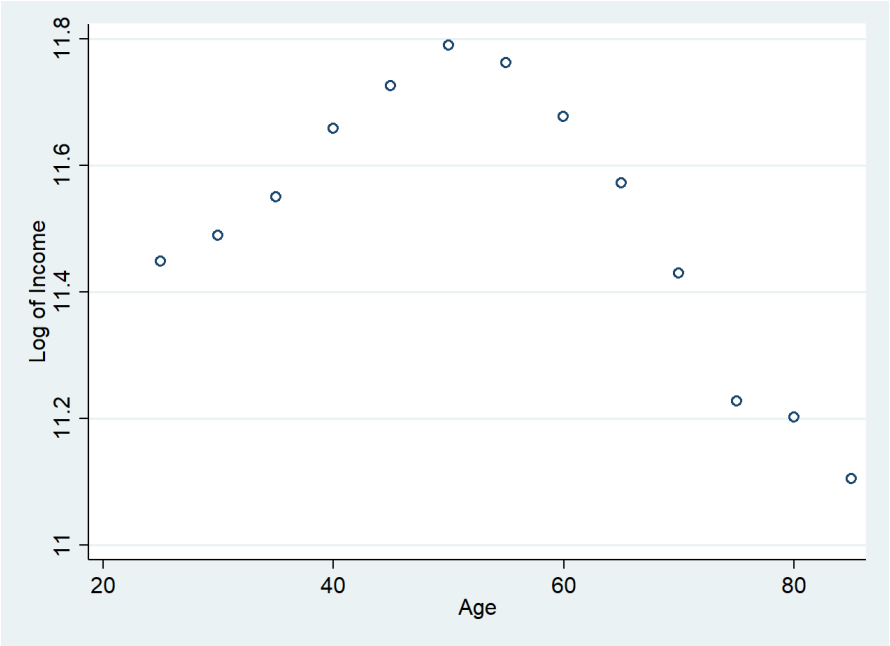

Figure 15: Life-cycle Average Income by Cohort from Townsend Thai Data in 2005-2014

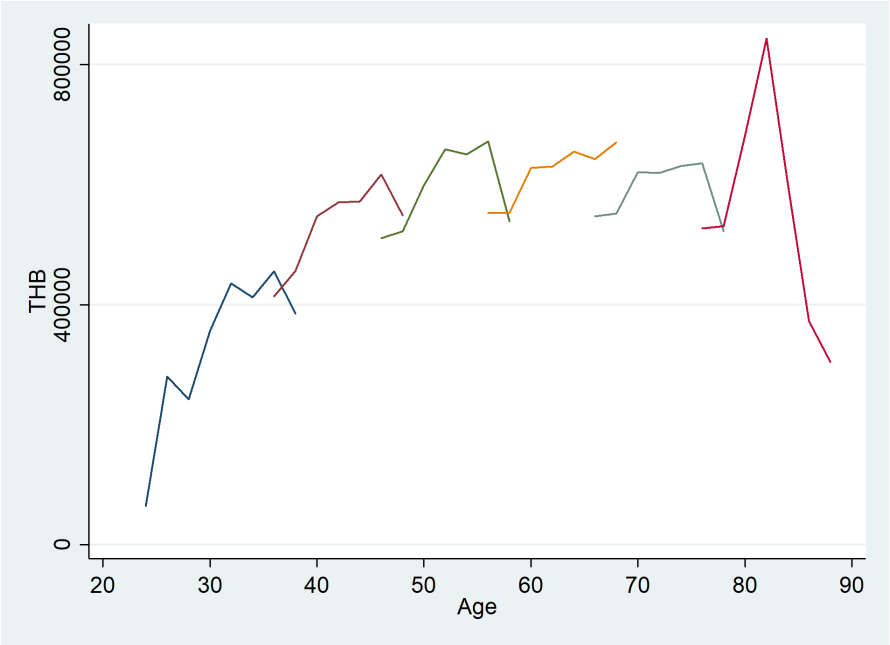

Figure 16: Life-cycle Average Assets by Cohort from Townsend Thai Data in 2005-2014

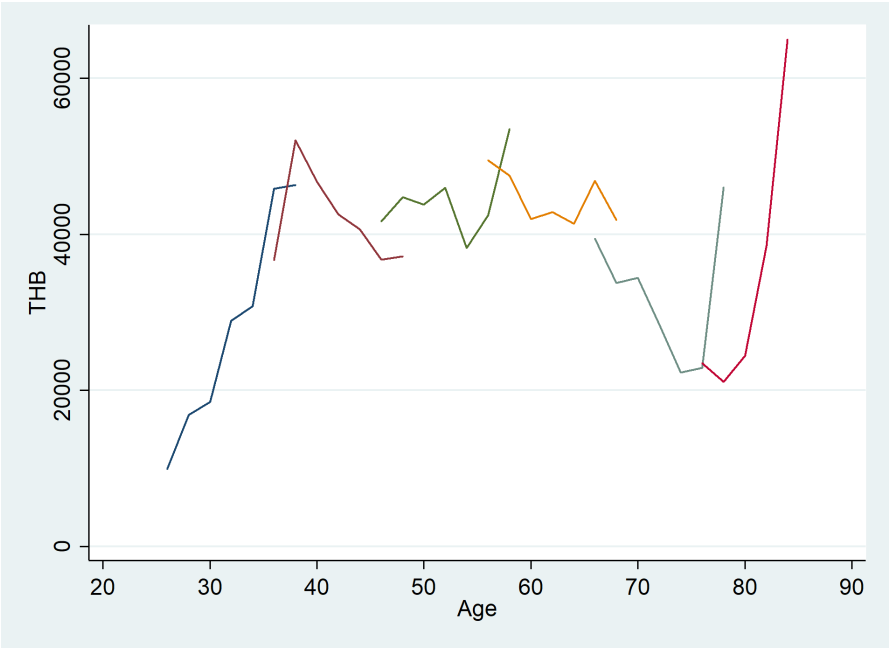

Figure 17: Life-cycle Average Personal Vehicle Holdings by Cohort from Townsend Thai Data in 2005-2014

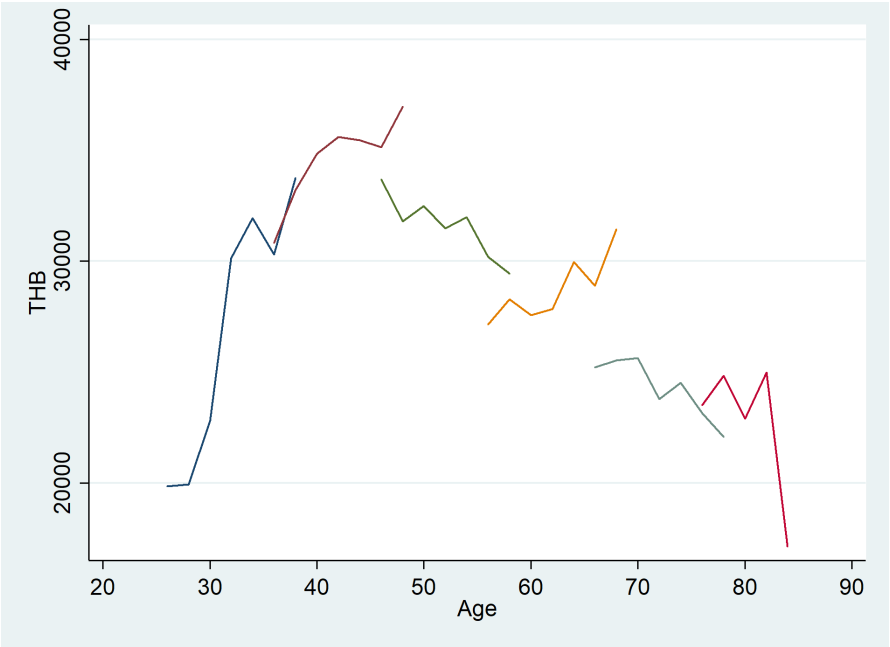

Figure 18: Life-cycle Average Non-durable Consumption by Cohort from Townsend Thai Data in 2005-2014 (Monthly Value)

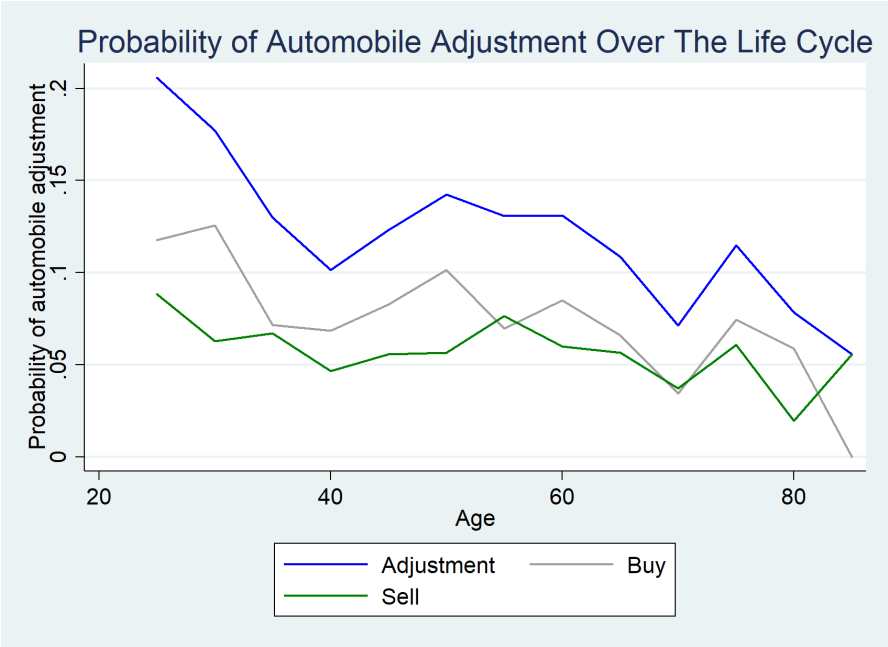

Figure 19: Life-cycle Average Probability of Adjustment by Age from Townsend Thai Data in 2005-2014

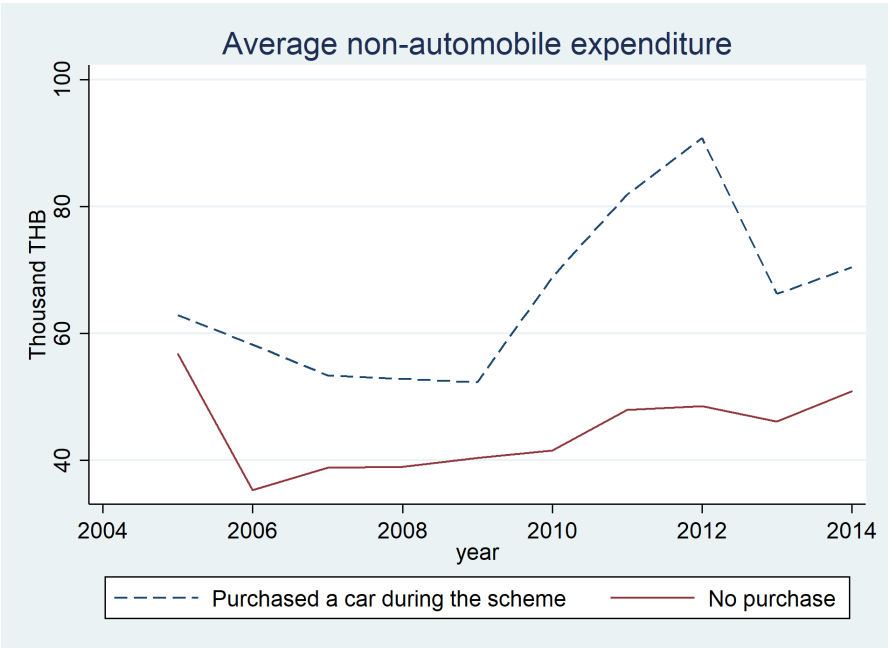

Figure 20: Average Non-durable Spending by Policy Participation From Townsend Thai Data in 2005 to 2014

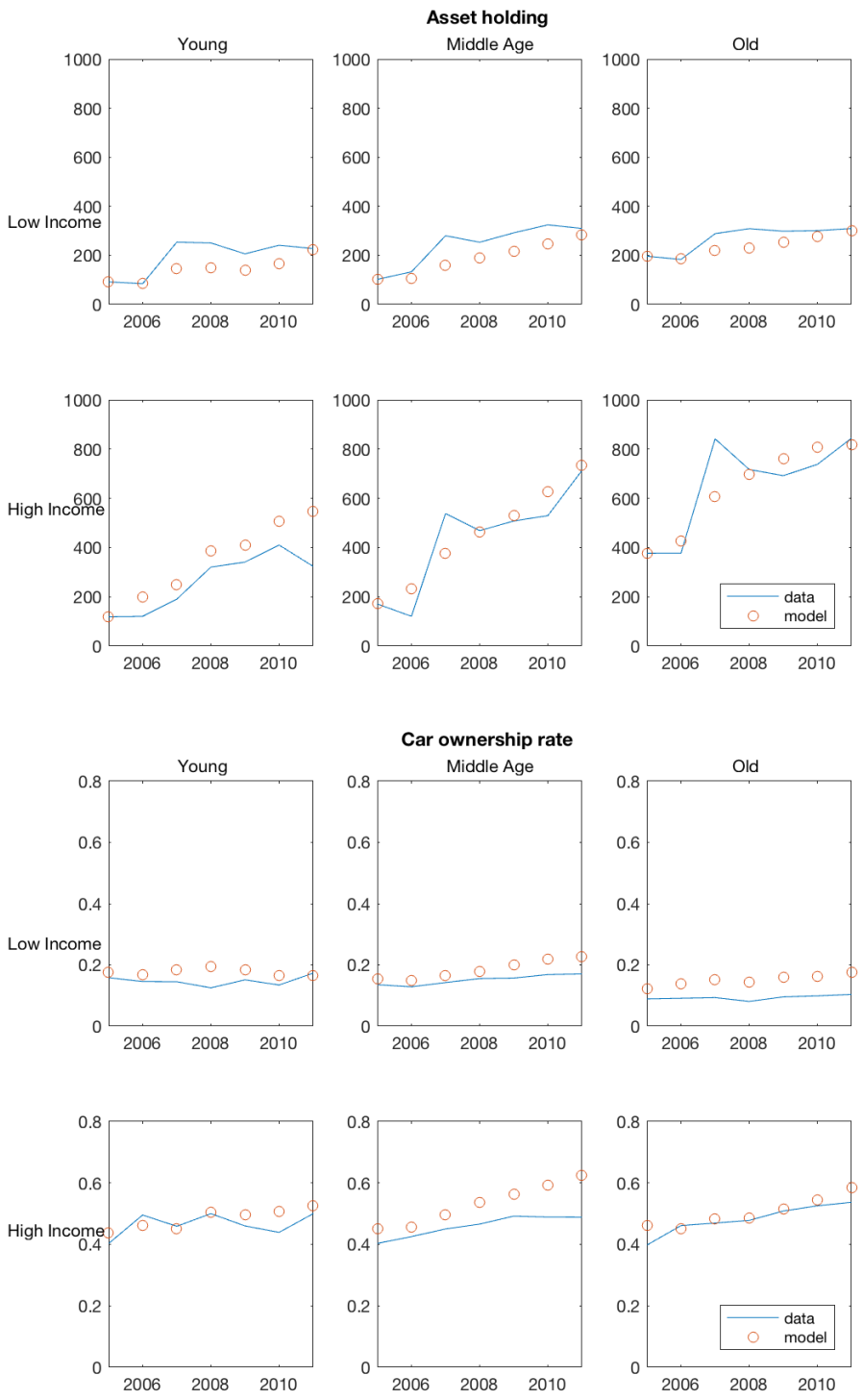

Figure 21: Targeted Moments: Assets and Fractions of Car Owners

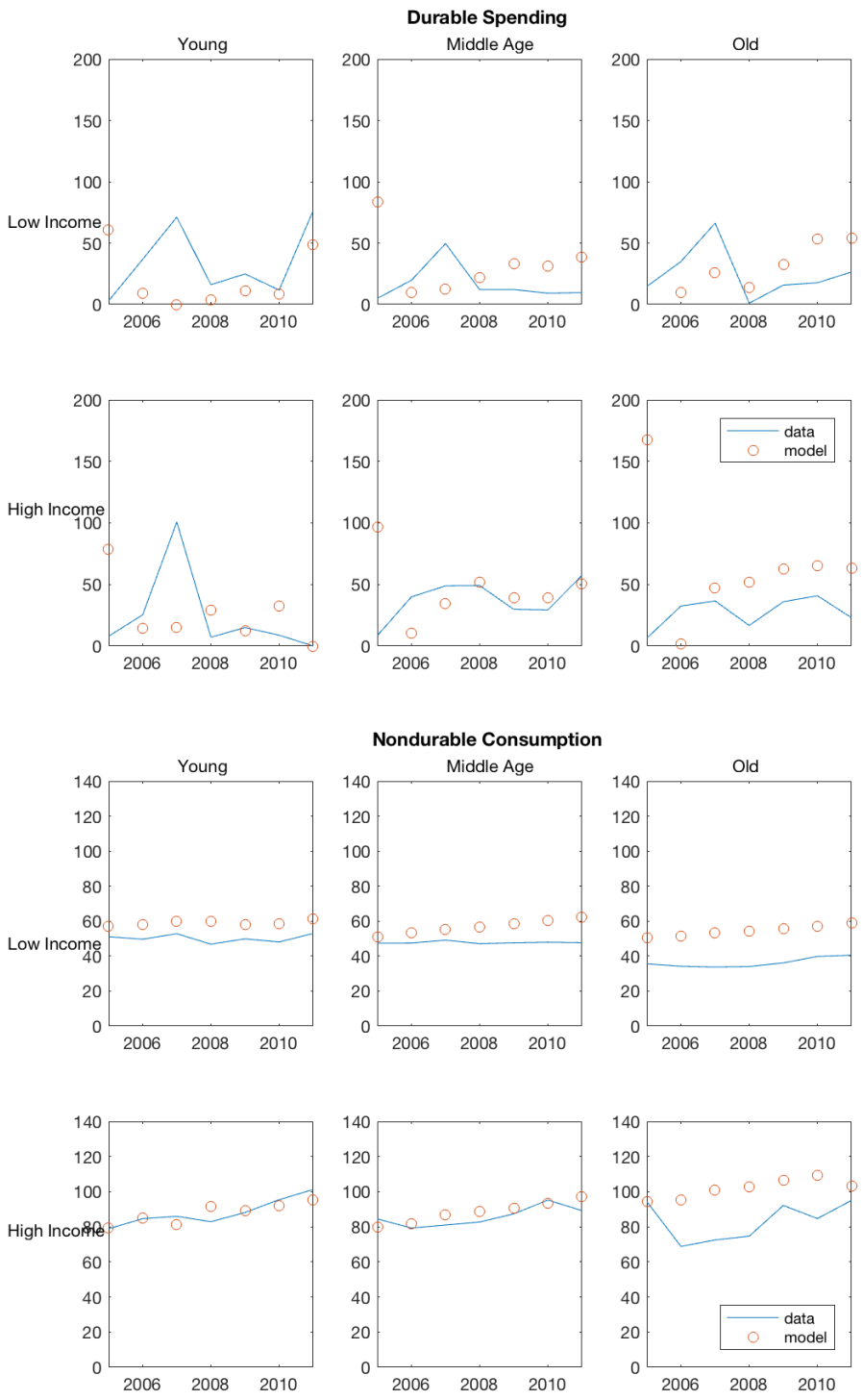

Figure 22: Targeted Moments: Durable and Non-durable Spending

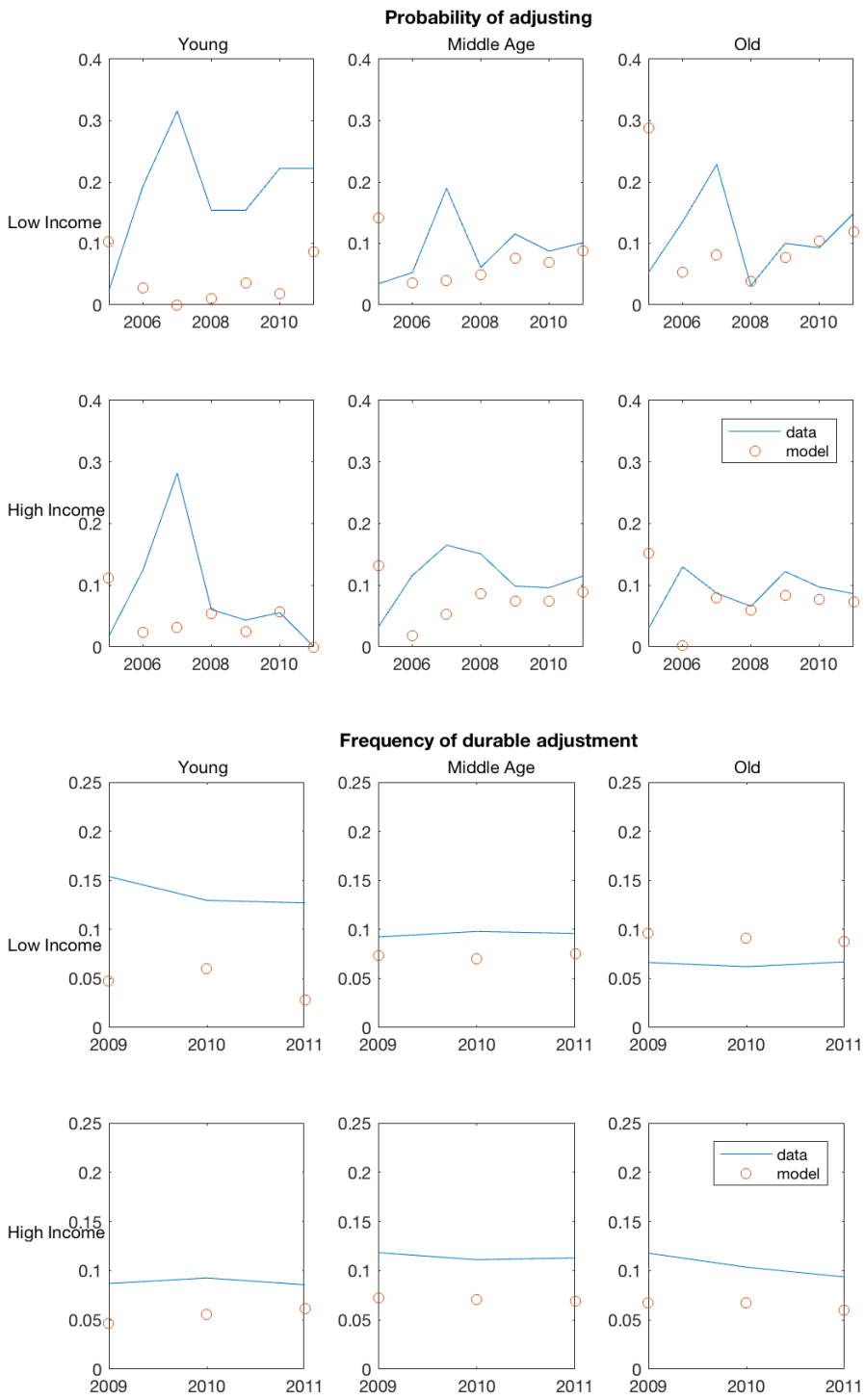

Figure 23: Targeted Moments: Frequency of Adjustment and Probability of Adjustment

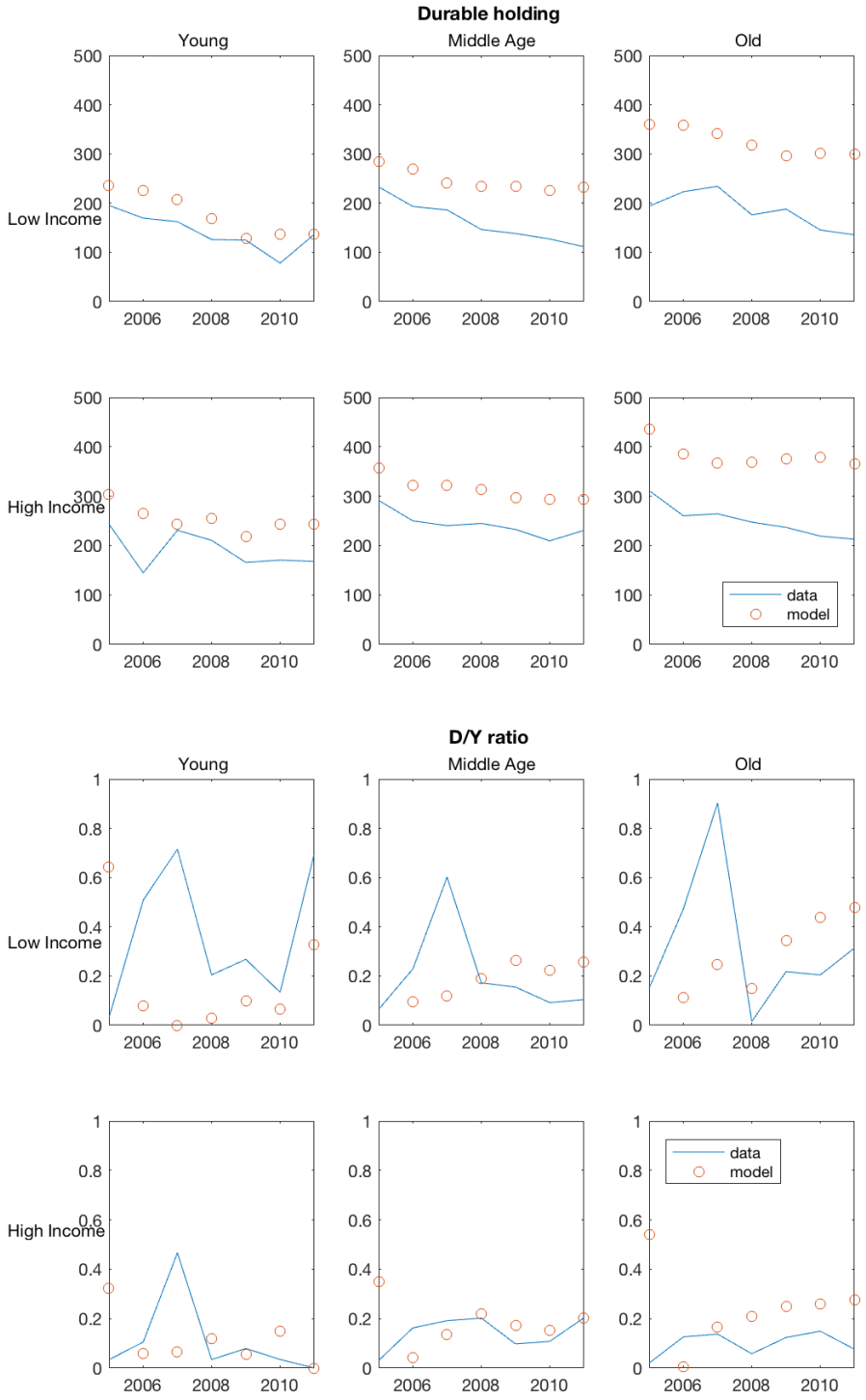

Figure 24: Non-targeted moments: Durable Stocks and Durable Spending to Income Ratio

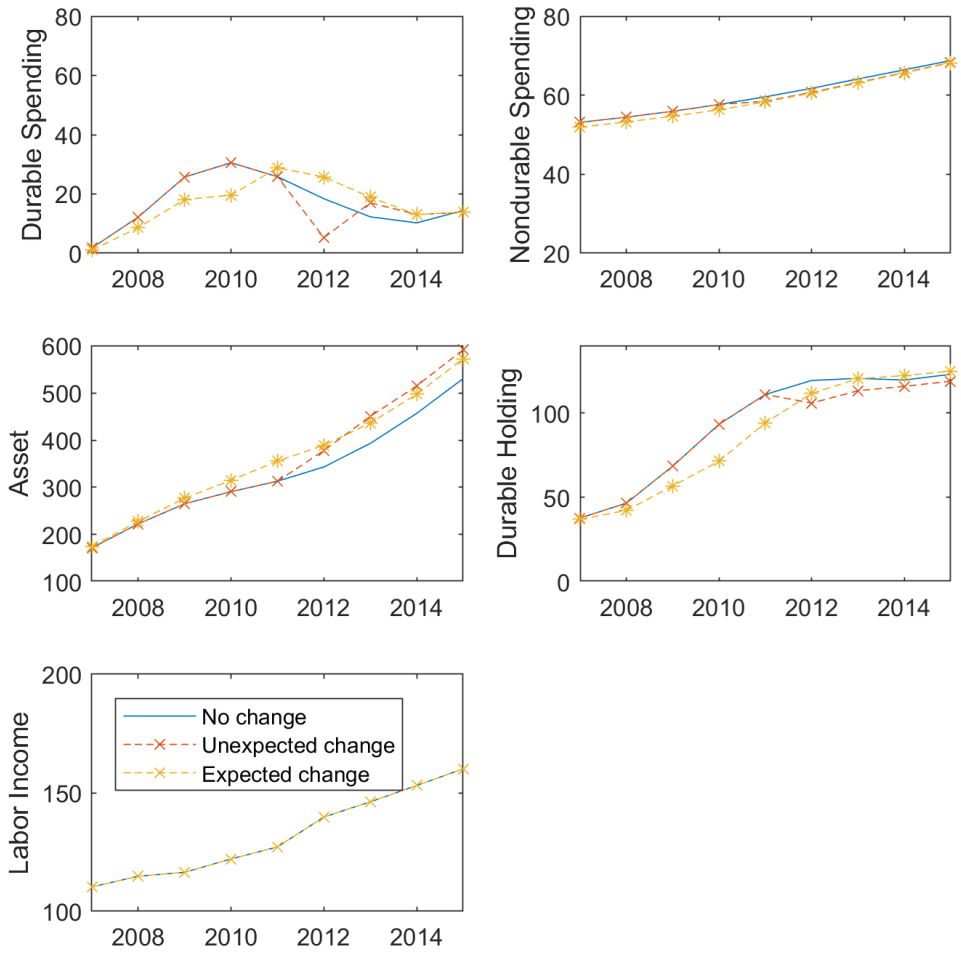

Figure 25: Policy Experiment with a Change in Asset Returns from 5% to 5.5%

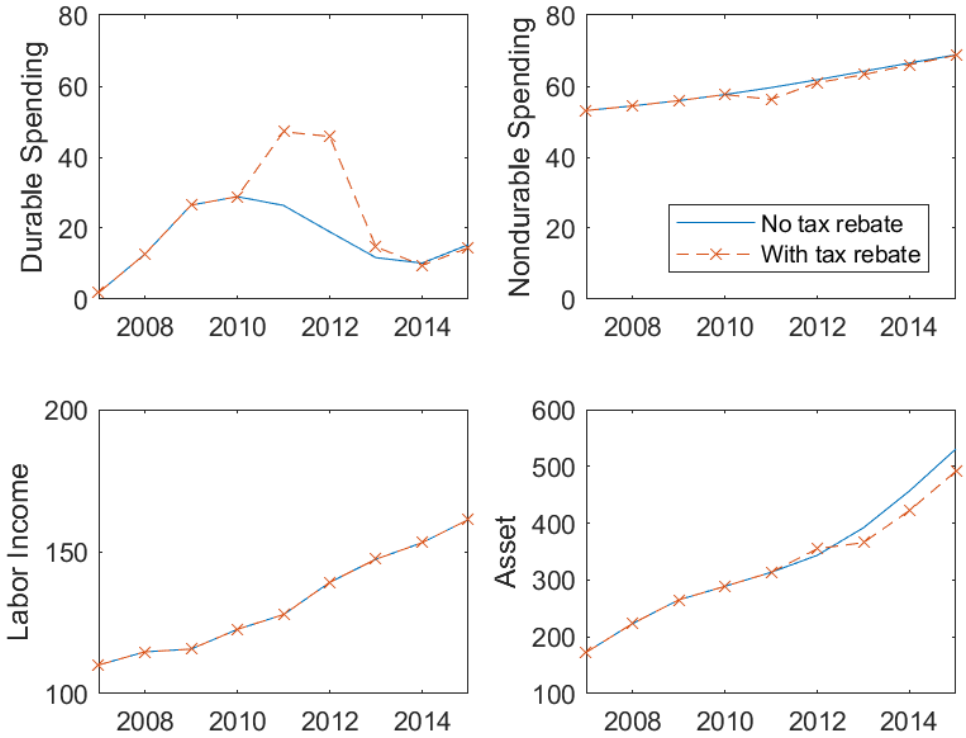

Figure 26: Policy Experiments: Car Tax Break in 2011-2012

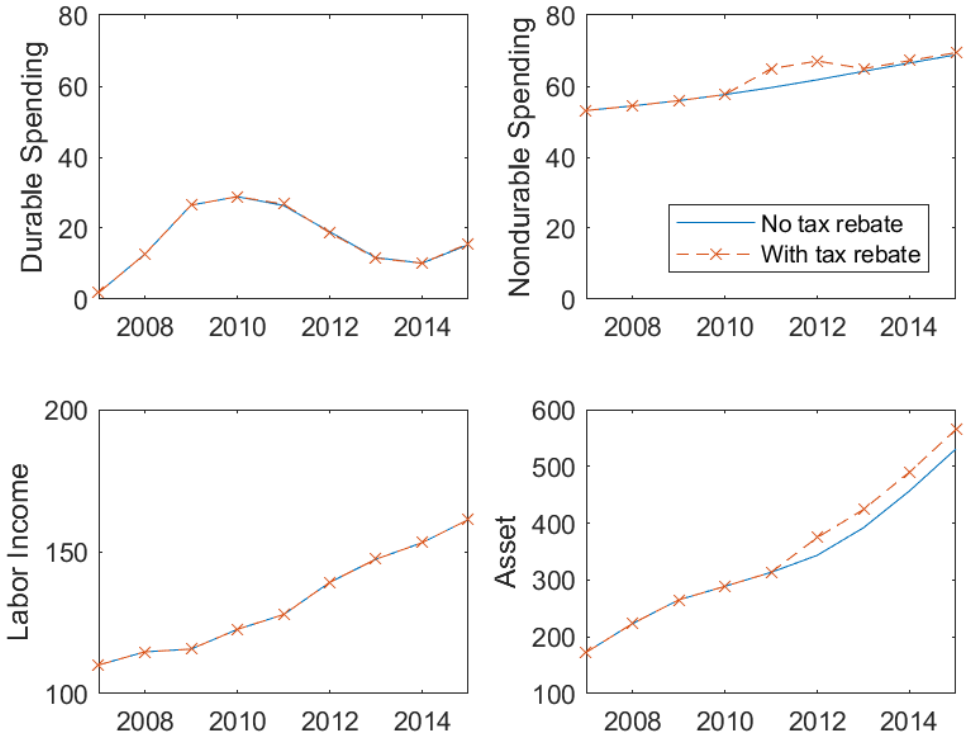

Figure 27: Policy experiment: Non-durable Consumption Tax Break in 2011-2012

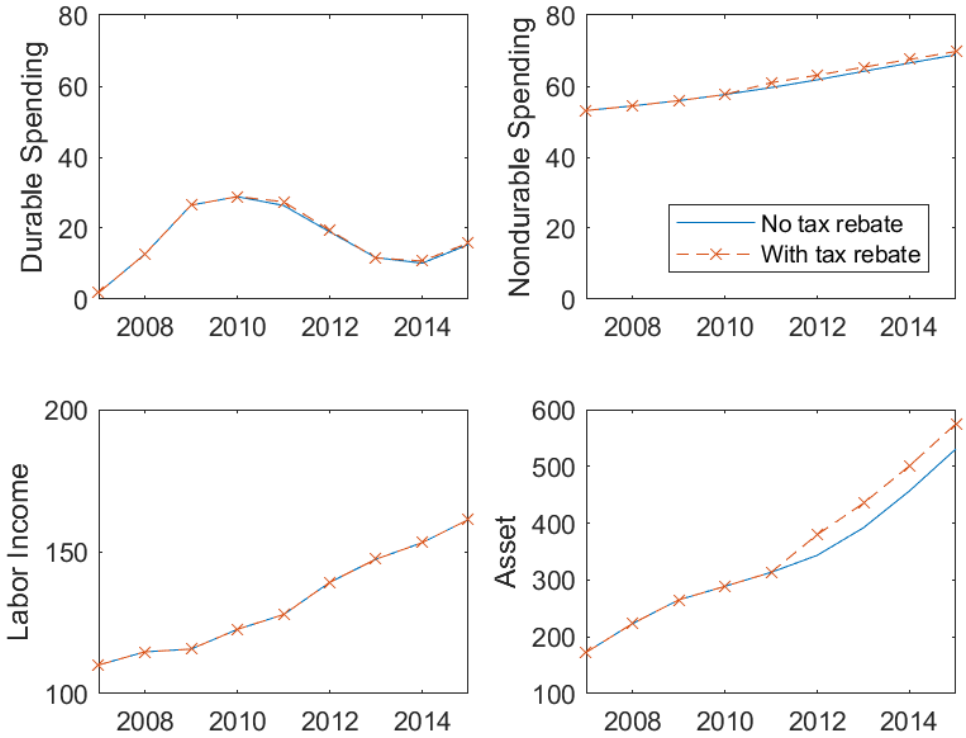

Figure 28: Policy Experiment: Income Tax Break in 2011-2012

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